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MILITARY SPECIFICATION

GYROSCOPES, REMOTE CONTROL (COMMAND) GUIDANCE SYSTEMS, GENERAL SPECIFICATION FOR

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the general requirements for the design and manufacture of gyroscopes for command guided missiles.

1.2 Classification. Gyroscopes covered by this specification shall be of the following types as specified:

Type I	-	Free gyroscope
Type II	-	Rate gyroscope
Type III	-	Vertical gyroscope
Type IV	-	Directional gyroscope
Type V	-	Rate integrating gyroscope

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-S-5002	Surface Treatments and Metallic Coatings for Metal Surfaces of Weapons Systems

FSC 1420

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STANDARDS

Military

	MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
	MIL-STD-129	Marking for Shipment and Storage
	MIL-STD-130	Identification Marking of U.S. Military Property
	MIL-STD-202	Test Methods for Electronic and Electrical Components Parts
	MIL-STD-210	Climatic Extremes for Military Equipment
	MIL-STD-446	Environments for Electronic Parts, Tubes and Solid State Devices
*	MIL-STD-454	Standard General Requirements for Electronic Equipment
*	MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
*	MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
*	MIL-STD-721	Definitions of Effectiveness Terms For Reliability, Maintainability, Human Factors, and Safety
*	MIL-STD-810	Environmental Test Methods
*	MIL-STD-831	Test Reports, Preparation of

* MANUALS

Military

Defense Standardization Manual 4120.3-M: Standardization Policies, Procedures and Instructions

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(Copies of Specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

* Aerospace Industries Association:

* EETC Report No. 5 (Revision A) - Standard Gyro Terminology, dated June 1965.

* (Applications for copies should be addressed to the National Standards Association, Inc., 1321 Fourteenth Street, N.W., Washington, D. C. 20005).

* The Institute of Electrical and Electronic Engineers, Inc.:

* IEEE Standards Publications:

* No. 292 (July 1969) - Specification Format for Single-Degree-of-Freedom Spring-Restrained Rate Gyros

* No. 293 (July 1969) - Test Procedure for Single-Degree-of-Freedom Spring-Restrained Rate Gyros

* (Applications for copies should be addressed to The Institute of Electrical and Electronic Engineers, Inc., 345 East 47 Street, New York, N. Y. 10017).

* Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction. Unless otherwise specified (see 6.2), a preproduction sample of the gyroscopes to be furnished under the terms of the contract or purchase order, shall be delivered to the testing activity designated in the contract or purchase order for preproduction testing.

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* 3.2 Detail specification. The individual item requirements shall be as specified herein and in accordance with the applicable detail specification. In the event of conflict between this general specification and the detail specification, the latter shall govern.

* 3.2.1 Preparation guides (rate gyros). IEEE Standards Publication Numbers 292 and 293 shall be used as guides in the development and preparation of associated detail specifications applicable to single-degree-of-freedom spring-restrained rate gyroscopes. Defense Standardization Manual 4120.3-M is mandatory for establishing policy and format of detail specifications.

3.3 Materials. The selection of materials for the design shall be made with special attention to the stringent requirements of thermal coefficient of linear expansion, thermal conductivity, specific heat, strength, hardness, density, and magnetic permeability.

3.3.1 Critical and strategic materials. Materials shall be selected on the basis of suitability and relative availability, taking into account the additional restrictions created during time of national emergency. Subject to satisfactory gyroscope design and where alternate materials with suitable physical characteristics are available, the design shall incorporate the least critical and strategic materials.

3.3.2 Elastomeric materials. Elastomeric components shall be fabricated from materials having maximum practicable ozone and aging resistance consistent with performance requirements and applicable specifications.

3.3.3 Electrolytic corrosion of metal parts. The use of dissimilar metals in immediate contact which will tend to accelerate electrolytic corrosion in the presence of moisture shall be avoided where design and performance considerations are not compromised. Where the use of such metals in immediate contact cannot be avoided, and where the interface between the metals is not wholly contained within a hermetically sealed enclosure, proper protection or insulation of such parts shall be provided.

3.3.4 Stress corrosion. Materials, techniques, and processes shall be selected and employed with regard to heat treatment procedures, corrosion protection, finish, assembly, and installation

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such that sustained residual surface tensile stresses, stress concentrations, the hazard of stress corrosion, cracking, and hydrogen embrittlement are minimized.

3.4 Design and construction. Design and construction of the gyroscopes shall be provided for the maximum practicable structural integrity combined with physical compatibility of the unit with its associated equipment.

3.4.1 Characteristics. Characteristics of the gyroscope design shall be such as to enable the gyroscope to satisfactorily perform its designated function as an essential component of the guided missile control system.

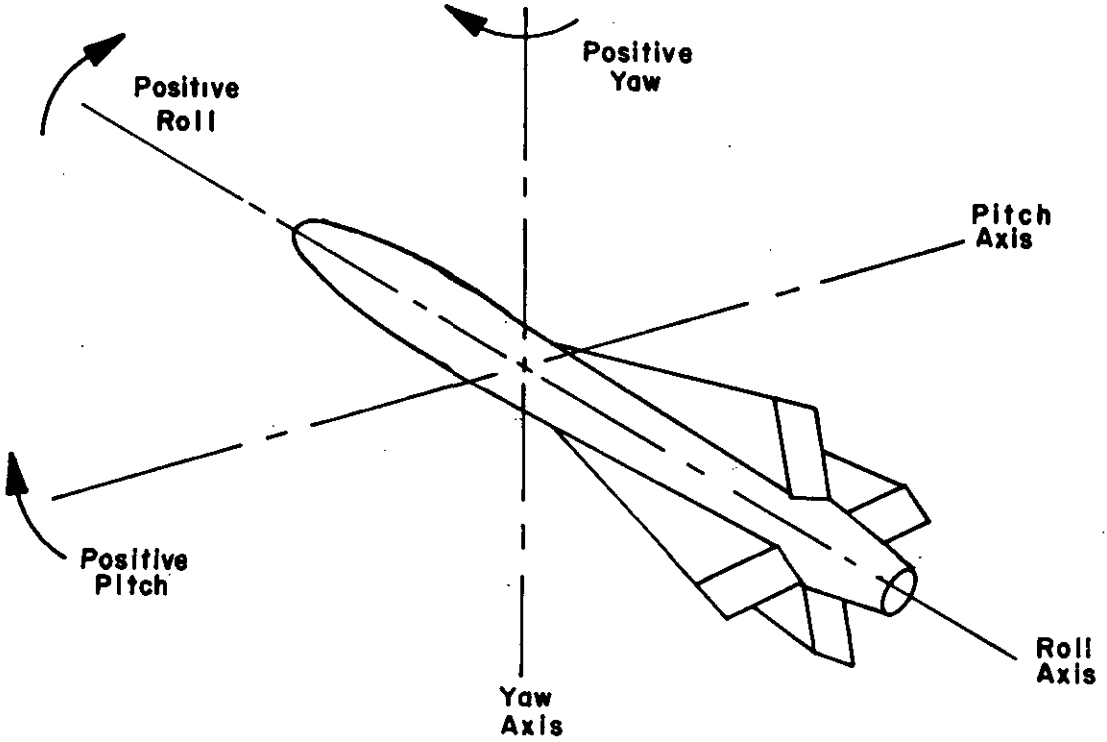
3.4.2 Simplicity. The design shall provide for a maximum of simplicity commensurate with all necessary specification performance criteria.

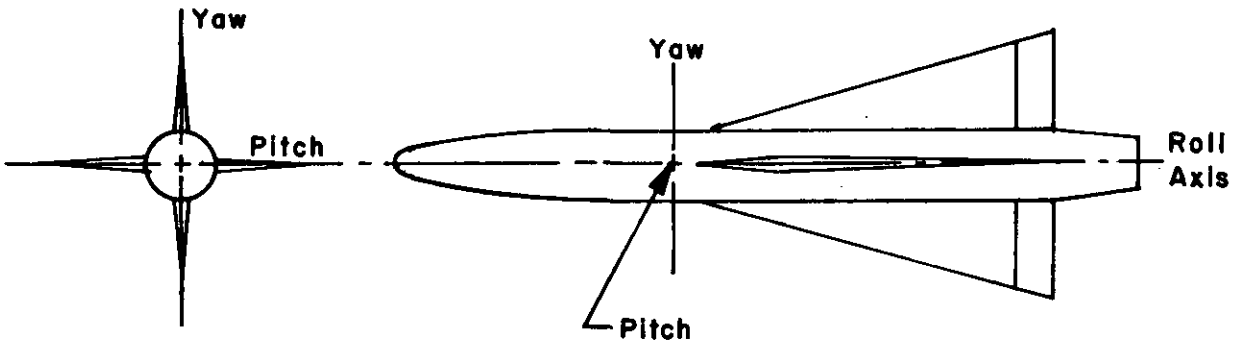
3.4.3 Mounting provisions. The mounting of the gyroscope is the interface of the specified gyroscope structure and the missile of which it is to become a functional part. Critical features of the mounting geometry shall be as specified herein and in the detail specification.

3.4.3.1 Detail drawing of mounting surface. The procuring activity shall provide a detailed dimensioned drawing of the surface to which the gyroscope must mount, showing all hole sizes, locations, and tolerances on these dimensions, to specify to a sufficient extent the mounting which must be provided on the gyroscope. The projected envelope of maximum dimensions (see 3.10) should be shown on this drawing.

* 3.4.3.2 Reference axis identification. Two of the three orthogonal principal reference axes of the missile, or gyroscope mounting platform, shall be clearly shown, located, and identified on the system drawings as the pitch, yaw, and roll axes in accordance with the definitions of figure 1. The polarities of the pitch, yaw, and roll axes shall be as represented in figure 1. If the third principal axis is not perpendicular to the plane of the section depicted on the drawing, such drawings as necessary to properly define all three axes should be provided.

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NOTE: Standard polarities (from rear of missile):

Positive Pitch - Nose up
Positive Yaw - Nose right
Positive Roll - Clockwise

* Fig. 1 MISSILE PRINCIPAL REFERENCE AXES

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3.4.3.3 Reference axis alignment provisions. The gyroscope mounting shall be designed to facilitate and guarantee a means of proper alignment of the gyroscope principal reference axes with the principal reference axes of the missile, or gyroscope mounting platform, and in a specific orientation sense (see 3.12.1).

3.4.3.4 Dimensional accuracy of mount. The dimensional accuracy of the gyroscope mounting shall be such that with the maximum possible misalignment of the gyroscope due to extremes of the mounting hole location and size tolerances, or due to extreme tolerance values of any other mounting technique, the gyroscope principal reference axis angular misalignment shall be less than a particular plus or minus value specified in the detail specification.

3.4.3.5 Rigidity of mount. The mounting arrangement of the gyroscope shall be such that under specified conditions of maximum sustained acceleration of the gyroscope, the angular deflection of the gyroscope principal reference axes shall be less than a particular plus or minus value specified in the detail specification.

3.4.4 Electrical connector. Unless otherwise specified (see 6.2), the gyroscope shall have a single electrical connector which shall accommodate all electrical connections required between the gyroscope and the outside system. This shall include provisions for gyroscope testing and monitoring connections when required. The connector shall conform to the applicable military specification set forth in the detail specification. When the gyroscope is required to be hermetically sealed (see 3.7.3), and applicable hermetically sealed connector shall be specified.

3.4.5 Surface finish. Unless otherwise specified (see 6.2), surface treatments of the gyroscope metal parts exposed to the environment (not contained within a hermetically sealed enclosure) shall be in accordance with MIL-S-5002.

3.4.6 System compatibility. The gyroscope shall be designed so as to be fully compatible with the missile systems power supplies. It shall be capable of accepting input information in the manner in which it is presented by other elements of the guidance and control system. It shall also yield its output information in a form which is readily utilized by other system elements. The normal operation of the gyroscope shall not interfere in any way with the successful operation of other components of the missile system.

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3.4.7 Corrosive fumes. Unless otherwise specified (see 6.2), the component materials shall not liberate, during the course of the gyroscope operation, any fumes or propellant combustion residue which may be deleterious to the gyroscope components and surrounding assemblies. The components and materials utilized within an hermetically sealed case shall not liberate contaminating fumes which may be deleterious to critical gyroscope components to an extent that specified performance is impaired following an extended storage time period.

3.4.8 Explosive atmosphere. Unless otherwise specified (see 6.2), the gyroscope shall not initiate combustion of a proximal explosive atmosphere.

3.4.9 Operational life. A minimum operational lifetime of the gyroscope shall be specified in the detail specification. This lifetime may be expressed in total operating hours, for the continuous drive type, or in terms of the number of drive and coast cycles for the drive and coast type. The anticipated maximum flight time of the missile as well as the anticipated aggregate amount of the test time or cycles of the gyroscope must be taken into account.

3.4.10 Mean time between failure (MTBF). The mean time between failure design objective shall be met with a certain minimum confidence level as specified in the detail specification.

* 3.4.11 Production. The gyroscope shall be readily manufactured by the techniques of mass production with no sacrifice in unit performance with respect to that given by the prototype model. To the extent practicable, the detail design shall be such that established processing equipment and techniques may be employed. The design shall require mechanical finishes and tolerances no more stringent than necessary to assure operation at the established reliability level throughout the test, storage, and operating life of the gyroscope.

3.5 General functional characteristics (all gyroscope types). The gyroscope shall be designed so as to maximize its flexibility of application to as wide a range of diverse applications within the scope of its type as is possible.

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3.5.1 Servicing and testing. The gyroscope shall be designed so as to require no servicing during its operational lifetime status in the missile. If this requirement cannot be fulfilled within the limits of design practicality, provisions shall be made to provide ready accessibility to the portions of the gyroscope where service is required. Provisions for gyroscope operational testing shall be made when operational testing is specified as a requirement of the detail specification.

* 3.5.2 Insulation resistance. The insulation resistance of the respective dielectric barriers between physically adjacent, but electrically distinct electric components and systems, or structural electrically conductive portions of the gyroscope, shall not be less than a specified value in ohms for each barrier in accordance with the detail specifications.

* 3.5.3 Warm-up time. The gyroscope shall attain its specified performance within a specified maximum time interval following its energization under specified operating conditions.

3.5.4 Rotor drive types. The gyroscope rotor drive scheme to be used shall be specified as a continuous rotor drive lasting the duration of the missile mission, or as a drive and coast method of rotor drive, with an initial rapid spin-up before or at missile launch, and a coast down interval which is greater than or equal to the duration of the missile mission. A combination of these schemes may be used if neither one of the alternate schemes can alone satisfy the rotor drive requirements in the following subparagraphs. The rotor drive types are listed as follows:

Type I	-	Induction motor
Type II	-	Hysteresis motor
Type III	-	D.C. motor
Type IV	-	Other

3.5.4.1 Maximum run-up time. The maximum time elapsed from the initiation of gyroscope rotor spin to the attainment of design rotor speed shall be as specified in the detail specification.

* 3.5.4.2 Electrical excitation - rotor drive type I or type II. The excitation for a type I or type II rotor drive shall conform to the detail specification in regard to the following requirements:

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a. Nominal RMS voltage value

b. Allowable voltage deviation

c. Nominal excitation frequency

d. Allowable frequency deviation

e. Number of phases

f. Maximum instantaneous current (starting)

g. Maximum RMS current (sustaining)

h. Excitation wave shape

3.5.4.3 Electrical excitation - rotor drive type III. The electrical excitation for a type III rotor drive shall conform to the detail specification in regard to the following requirements:

a. Nominal D. C. voltage

b. Allowable voltage deviation

c. Maximum instantaneous current (starting)

d. Maximum sustaining current

3.5.4.4 Rotor drive type IV. All essential characteristics of the type IV rotor drive that relate to the power source and its control shall be specified in the detail specification.

3.5.4.4.1 Operation time. The minimum time interval following attainment of maximum rotor speed, throughout which the gyroscope must maintain the specified performance shall be specified in the detail specification.

* 3.5.5 Pick-offs. The angular position pick-off shall be a transducer which will uniquely convert the angular position variable into an electrical signal, either analog or digital, which is compatible with the gyroscope output information required by the remainder of the missile guidance and control system. All gyroscope types require at least one pick-off, however, a two degree of freedom gyroscope may require a pick-off on both output reference axes. The detail specification shall then provide requirements for both pick-off number 1 and number 2. When only one pick-off is required, or when the requirements for the two pick-offs are identical, the requirements for pick-off number 1 shall apply.

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3.5.5.1 Pick-off types. The pick-off shall be one of the following types, as specified:

Type I - Potentiometer (wirewound, film, or photoelectrically coupled)

Type II - Active AC (synchro resolver, linear variable differential transformer, etc.)

Type III - Other (commutator, limit switch pair, digital shaft-angle encoder, etc.)

* 3.5.5.2 Pick-off polarity. In type I pick-off types, the voltage from the pick-off signal line to the pick-off excitation line shall increase with positive input as defined in the basic diagram of the detail specification. In type II pick-off types, the voltage from the pick-off secondary line-to-return shall be in phase, within pick-off phase shift, with the voltage from the pick-off primary line-to-return for positive inputs as defined in the basic diagram of the detail specification. The positive output signal polarity of the type III pick-off shall be as specified in the detail specification.

* 3.5.5.3 Pick-off characteristics. Pick-off number 1 and pick-off number 2 shall be of types here designated and shall have the following characteristics.

* 3.5.5.3.1 Pick-off output nonlinearity. The pick-offs shall accurately indicate the angular position of the output axis with respect to the gyro case over a specified minimum magnitude of angular ranges from the nominal null position. The allowable deviation from the true angular position must be specified in volts, degrees, or percent of full scale for positive and negative angular positions. Positive and negative output angles need not be equal when specified.

3.5.5.3.2 Pick-off type I. The characteristics of a type I pick-off shall conform to the detail specification in regard to the following requirements:

a. Total resistance (nominal)

b. Allowable resistance deviation from nominal

c. Maximum continuous power dissipation

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* 3.5.5.3.2.1 Shorting strips. When shorting strips or equivalent means are required to maintain pick-off electrical continuity during gyroscope input overload conditions, or for angular displacement magnitudes greater than the corresponding specified pick-off output angle, but less than the corresponding angular freedom, it shall be so specified in the detail specification.

* 3.5.5.3.3 Pick-off type II. The characteristics of a type II pick-off shall conform to the detail specification in regard to the following requirements:

a. Nominal RMS excitation voltage
b. Allowable voltage deviation
c. Nominal excitation frequency
d. Allowable frequency deviation
e. Number of voltage phases
f. Maximum continuous excitation power
g. Maximum RMS output voltages
h. Maximum harmonic content

3.5.5.3.4 Pick-off type III. All essential characteristics of the type III pick-off that relate to the interface of the pick-off and the rest of the control system, including excitation, output, and polarity, shall be specified in the detail specification along with the particular design details of the pick-off.

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* 3.5.6 Torquers. When torquer or torquers (for two degree of freedom gyroscopes) are required, the torquer shall be a transducer which will uniquely convert the electrical signal input variable corresponding to a desired value of torque to an actual output torque applied to the rotor about the desired output axis. If the detail specification is for a two degree of freedom gyroscope and a torquer is required for each output axis, then the required information for both torquer number 1 and number 2 shall be provided. When only one torquer is required, or when the requirements for the two torquers are identical, the information for torquer number 1 shall apply.

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* 3.5.6.1 Torquer types. The torquer shall be one of the following types, as required by the procuring activity:

Type I	-	AC Torquer
Type II	-	DC Torquer
Type III	-	Other

* 3.5.6.2 Torquer polarity. The positive command signal input polarity of the type I torquer shall be a voltage of the same frequency as the excitation voltage and in time phase with the excitation voltage. The positive command signal input polarity of the type II torquer shall be a positive input voltage with respect to the reference potential. The positive command signal input polarity of the type III torquer shall be as specified in the detail specification.

3.5.6.3 Torquer number 1. The torquer number 1 shall be of a type here designated, and shall have the following characteristics:

3.5.6.3.1 Torquer type I. The characteristics of a type I torquer shall conform to the detail specification in regard to the following requirements:

a. Nominal RMS excitation voltage
b. Allowable voltage deviation
c. Nominal excitation frequency
d. Allowable frequency deviation
e. Number of phases
f. Maximum continuous input power

3.5.6.3.2 Torquer type II. The characteristics of a type II torquer shall conform to the detail specification in regard to the following requirements:

a. Maximum DC voltage
b. Maximum continuous input power
c. Maximum composite error

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* 3.5.6.3.2.1 Torquer specified scale factor. The scale factor of the torquer shall be as specified in the detail specification with allowance for gain adjustment of a torquer control amplifier where applicable.

3.5.6.3.3 Torquer type III. All essential characteristics of the type III torquer that relate to the interface of the torquer and the rest of the control system, including excitation, input, and polarity shall be specified in the detail specification along with the particular design details of the torquer.

3.5.6.4 Torquer number 2. The torquer number 2 shall be of a type here designated, and shall have the following characteristics.

3.5.6.4.1 Torquer type I. The characteristics of a type I torquer shall conform to the detail specification in regard to the following requirements:

a. Nominal RMS excitation voltage

b. Allowable voltage deviation

c. Nominal excitation frequency

d. Allowable frequency deviation

e. Number of phases

f. Maximum continuous input power

3.5.6.4.2 Torquer type II. The characteristics of a type II torquer shall conform to the detail specification in regard to the following requirements:

a. Maximum DC voltage

b. Maximum continuous input power

c. Maximum composite error

* 3.5.6.4.2.1 Torquer specified scale factor. The scale factor of the torquer shall be as specified in the detail specification with allowance for gain adjustment of a torquer control amplifier where applicable.

* 3.5.6.4.3 Torquer type III. All essential characteristics of the type III torquer that relate to the interface of the torquer and the

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rest of the control system, including excitation, input, and polarity shall be as specified in the detail specification along with the particular design details of the torquer.

* 3.5.7 Temperature control. Electric heaters may be used to realize specified performance and parameter stability in the face of the specified range of environments. When such heaters are required, the details of sensors, heater requirements and the voltage and power available for the excitation shall be as specified in the detail specification.

3.5.8 Special features. Any special features or functional requirements (such as gyro rotor braking, loss of power indication, caging or uncaging indicator switches, rotor rotation detectors, etc.) that may be required shall be specified in the detail specification.

3.6 Specific functional characteristics (by gyroscope type). The requirements of this paragraph apply only to specific gyroscope types as indicated in the respective subparagraphs 3.6.1, 3.6.2, etc.

3.6.1 Type I - free gyroscope. These requirements shall apply to the type I gyroscope only.

* 3.6.1.1 Reference axis identification. The two orthogonal gyroscope output reference axes (ORA) shall each be identified on the system drawings as the pitch, yaw, or roll axes in accordance with their relation to the missile principal reference axes of figure 1. The gyroscope spin reference axis (SRA) shall be identified on the system drawings as the pitch, yaw, or roll axis in similar fashion. This requirement is intended to establish the gyroscope orientation with respect to the missile only, and does not imply any restriction on the gyroscope design such as which gimbal shall be the inner gimbal, and which shall be the outer gimbal, or even if gimbals must be used at all.

* 3.6.1.2 Output axis gimbal freedom. The two ranges (positive and negative) constituting the output axis span of the spin axis about each of the output reference axes shall be specified for each of the output reference axes named in 3.6.1.1. The design of the free gyroscope shall be such that the output axis ranges will not permit gimbal lock to occur, if gimbals are used.

* 3.6.1.3 Output axis pick-offs. The output reference axes which require pick-offs shall be named by the gyroscope output reference

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axis for which they provide displacement angle signal. They shall also be associated with the appropriate pick-off number 1 or number 2 requirements of 3.5.5.3.

* 3.6.1.4 Torquers. The output axes which require torquers shall be named and associated with the appropriate torquer number 1 or number 2 requirements of 3.5.6.3.

* 3.6.1.4.1 Minimum range of torquing rate. A type II torquer shall be capable of applying a continuous range of torques such that the precession rate about the output axis orthogonal to the torquer axis may be varied in magnitude continuously throughout a range from zero to a specified maximum value. This minimum range of torquing rate shall be the same for both the plus and minus precession (angular freedom) ranges.

* 3.6.1.5 Caging mechanism. The type of caging mechanism shall be specified as either electrical, mechanical or electromechanical. Other pertinent design details such as uncaging upon energization (so that the gyroscope is caged during handling and shipment), shall be specified in the detail specification.

* 3.6.1.5.1 Caging mechanism performance. The caging mechanism shall conform to the detail specification.

* 3.6.1.6 Gyro scale factor. The gyro scale factor (see 6.3.10) shall be in accordance with the detail specification.

* 3.6.1.7 Performance. The free gyroscope shall conform to the following performance requirements as specified in the detail specification.

3.6.1.7.1 Maximum composite error (both axes). The composite error (see 6.3.16) of the gyroscope applicable to both output axes shall not exceed the specified maximum value.

* 3.6.1.7.2 Maximum cross-coupling error. Cross-coupling error for inputs about either input axis (IA) shall not exceed the specified maximum value.

* 3.6.1.7.3 Maximum electrical zero (at angular displacement nominal zero). The excited output voltage waveform for either pick-off when the gyroscope is caged (the spin axis is nominally coincident with the spin reference axis), shall not exceed a specified value.

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* 3.6.1.7.3.1 Maximum signal null. For the type II pick-off, the fundamental component (signal component) of the excited output voltage waveform for either pick-off shall not exceed a specified value.

* 3.6.1.7.4 Maximum caging error. The angular error between the gyro output axis and the output reference axis when the gyro is caged shall be within the limit specified in the detail specification.

* 3.6.1.7.5 Minimum nutation frequency. The nutation frequency (see 6.3.17) shall not fall below a specified minimum value.

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* 3.6.1.7.6 Maximum free drift rate. The free drift rate shall not exceed the specified maximum value.

* 3.6.1.7.7 Maximum systematic drift rate (with specified conditions). The systematic drift rate (see 6.3.18) of the gyroscope when subjected to specified dynamic environment conditions shall not exceed the specified maximum value.

* 3.6.1.7.8 Maximum random drift rate. The random drift rate of the gyroscope, when subjected to specified dynamic environments, shall not exceed the specified maximum value.

* 3.6.1.7.9 Acceleration - sensitive drift rate. The drift rate caused by accelerations acting on the mass unbalance along any principle axis or gimbal shall not exceed the specified maximum value.

* 3.6.1.8 Polarity. The polarity of the positive electrical output signal of the pick-off and the positive electrical input command signal of the torquer are given in 3.6.1.8.1 and 3.6.1.8.2.

* 3.6.1.8.1 Pick-offs. The polarity of each pick-off shall be such that the component of spin axis angular displacement along the indicated output axis is a positive electrical signal when the missile has under-gone a positive angular displacement about the corresponding missile axis (see figure 1).

3.6.1.8.2 Torquers. The polarity of each torquer shall be such that when a positive command input electrical signal is applied, it produces an angular precession of the spin axis about the orthogonal reference axis corresponding to a positive pitch, yaw, or roll angular displacement (see figure 1).

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3.6.2 Type II - rate gyroscopes. These requirements shall apply to the type II gyroscopes only.

* 3.6.2.1 Reference axis identification. The gyroscope output axis or OA shall be identified on the detail system drawings as the pitch, yaw, or roll axis in accordance with its relation to the missile principal reference axes of figure 1. The gyroscope spin reference axis (SRA) and input reference axis (IRA) shall be identified on the detail system drawings as the pitch, yaw, or roll axes in similar fashion.

* 3.6.2.2 Input limit. The input limit in both positive and negative directions shall be within the performance requirements specified in the detailed specification.

* 3.6.2.3 Mechanical overload. For a specified time duration, the rate gyroscope shall be capable of withstanding an angular rotation rate about the input reference axis (IRA) of either polarity and a maximum magnitude specified in the detail specification. It shall suffer no damage or subsequent impairment of performance.

* 3.6.2.3.1 Limit stops. The output axis of the gyro shall be mechanically limited to an input limit value as specified in the detail specification.

* 3.6.2.4 Gimbal (self test) torquer. When an output axis torquer is required, it shall be so specified and the torquer number 1 requirements shall apply (see 3.5.6.3).

3.6.2.4.1 Minimum range of equivalent input rate. A type II torquer shall be capable of applying a continuous range of torques of either polarity such that any magnitude of equivalent input angular rotation rate about the input reference axis may be achieved. The minimum required range of equivalent input rate for either polarity shall be specified in degrees per unit time.

* 3.6.2.5 Gyro scale factor. The scale factor (see 6.3.10) shall conform to the detail specification.

3.6.2.6 Performance. The rate gyroscope shall conform to the following performance requirements as specified in the detail specification.

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3.6.2.6.1 Minimum undamped natural frequency. The minimum undamped natural frequency for the rate gyroscope (see 6.3.12) shall be a specified value which shall apply for the full range of operational environment conditions.

3.6.2.6.2 Nominal damping ratio. The nominal damping ratio (see 6.3.13) and appropriate tolerance limits for damping ratio variation under the full range of operational environment conditions shall be specified in the detail specification.

3.6.2.6.3 Maximum composite error. The composite error of the rate gyroscope shall not exceed a specified maximum value. Unless otherwise specified (see 6.2), the rate gyroscope shall also conform separately to the requirements of the following components of the composite error:

a. Maximum threshold

b. Maximum resolution

3.6.2.6.4 Maximum electrical null (at zero input angular rate). The excited output voltage waveform when there is no angular rotation rate of the gyroscope about the input reference axis (IRA) shall not exceed a specified RMS value.

3.6.2.6.4.1 Maximum signal null (at zero input angular rate). For the type II pick-off, fundamental component (signal component) of the excited pick-off output voltage waveform when there is no angular rotation rate of the gyroscope about the input reference axis (IRA), shall not exceed a specified RMS value.

* 3.6.2.6.5 Zero offset. The maximum gyro output signal with no rate applied about the input axis shall remain within the limits specified in the detailed specification.

* 3.6.2.6.6 Acceleration-sensitive drift rate. The acceleration-sensitive drift rate (see 6.3.14) shall not exceed the specified maximum value.

* 3.6.2.6.7 Acceleration-squared-sensitive drift rate. The acceleration-squared-sensitive drift rate (see 6.3.14) shall not exceed the specified maximum value.

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3.6.2.6.8 Rate sensitivity about the spin reference axis (SRA). Unless otherwise specified by the procuring activity (see 6.2), the maximum rate sensitivity about the spin reference axis (SRA) under various conditions of gyroscope angular rotation rate about the input reference axis (IRA), shall be specified as follows.

3.6.2.6.8.1 Zero rate input about the input reference axis (IRA). The rate gyroscope shall have less than a specified maximum sensitivity to angular rotation rates about the spin reference axis (SRA) with zero angular rotation rate of the gyroscope about the input reference axis (IRA).

3.6.2.6.8.2 Specified rate input about the input reference axis (IRA). The rate gyroscope shall have less than a specified maximum sensitivity to angular rotation rates about the spin reference axis (SRA) when a specified angular rotation rate is applied about the input reference axis (IRA). This angular rotation rate about the input reference axis (IRA) shall be at least one half the maximum input range.

3.6.2.6.9 Rate sensitivity about the output axis (OA). The rate gyroscope shall have less than a specified maximum sensitivity to angular rotation rates of the gyroscope about the output axis with zero angular rotation rate of the gyroscope about both the input reference axis (IRA) and the spin reference axis (SRA).

3.6.2.6.10 Angular acceleration sensitivity about the output axis (OA). The rate gyroscope shall have less than a specified maximum sensitivity to angular acceleration of the gyroscope about the output axis with zero angular rotation rate of the gyroscope about both the input reference axis (IRA) and the spin reference axis (SRA).

3.6.2.7 Polarity. The polarity of the positive electrical output signal of the pick-off and the positive electrical input command signal of the torquer are given as follows.

* 3.6.2.7.1 Pick-off. The polarity of the pick-off shall be such that the indicated angular rotation rate of the gyroscope about the input reference axis (IRA)(where the IRA coincides with a missile principal reference axis) shall be a positive electrical signal as defined in the basic diagram of the detail specification.

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* 3.6.2.7.2 Torquer. The polarity of the torquer shall be such that when a positive command input electrical signal is applied, it produces a positive signal at the pick-off.

3.6.3 Type III - vertical gyroscope. These requirements shall apply to the type III gyroscope only.

* 3.6.3.1 Reference axis identification. The two orthogonal gyroscope output axes shall be the pitch and roll axes. The spin reference axis (SRA) shall be the yaw axis. The gyroscope orientation with respect to the missile is fixed by its type (vertical) designation. These gyroscope principal reference axes are in correspondence with the missile principal reference axes of figure 1.

* 3.6.3.2 Gimbal freedom. The maximum angular displacement of the inner gimbal shall be as specified in the detail specification. The outer gimbal shall have unrestricted freedom unless otherwise specified.

* 3.6.3.3 Output axis pick-offs. The output axes which require pick-offs shall be named by the gyroscope output axis for which they provide displacement angle signal, and associated with the appropriate pick-off number 1 or number 2 requirements of 3.5.5.3.

* 3.6.3.3.1 Polarity. The polarity of the pick-off outputs shall be coincident with the corresponding missile angular displacement polarity as defined in figure 1.

3.6.3.4 Spin axis erection system. The vertical gyroscope shall incorporate a means for maintaining the rotor spin axis in a nominally vertical position, with an erection reference as specified below.

* 3.6.3.4.1 Erection reference. The detail specification shall establish whether the erection (see 6.3.19) reference line shall be the vertical defined by the yaw principal reference axis of the missile or the local vertical defined by the gravity field of the Earth (the "local vertical-referenced" vertical gyroscope).

* 3.6.3.4.1.1 Erection sensor operational range. The operational range of the erection sensors used for gravity erected vertical gyroscopes shall be such that the proper signal polarity (and amplitude, is proportional erection is used) shall be maintained throughout the missile attitude ranges specified in the detail specification.

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3.6.3.4.2 Excitation. The nominal RMS voltage, frequency, and continuous power available for excitation of both reference sensors shall be specified in the detail specification.

3.6.3.4.3 Erection rate. The nominal erection rate(s) at rotor design speed and the maximum plus or minus percentage variation of this erection rate from the nominal value shall be specified in the detail specification.

* 3.6.3.4.4 Erection accuracy. The vertical gyroscope shall satisfy or exceed a specified minimum erection accuracy for both output axes.

* 3.6.3.5 Caging. When caging (see 6.3.20) is required (see 6.2), the excitation voltage and power available for the caging mechanism, and such pertinent performance parameters as maximum caging and uncaging time shall be specified.

* 3.6.3.6 Gyro scale factor. The scale factor (see 6.3.10) of the gyroscope shall be in accordance with the detail specification.

3.6.3.7 Performance. The vertical gyroscope shall conform to the following performance requirements as specified in the detail specification.

* 3.6.3.7.1 Maximum composite error (both axes). The composite (see 6.3.16) of the gyroscope applicable to both output axes shall not exceed the specified maximum value.

* 3.6.3.7.2 Maximum cross-coupling error. The cross-coupling error for either output axis shall not exceed the specified maximum value.

* 3.6.3.7.3 Maximum electrical zero (at angular displacement nominal zero). The excited output voltage for either pick-off when the rotor spin axis is vertical and aligned with the spin reference axis (SRA), shall not exceed a specified value.

* 3.6.3.7.4 Maximum erection/caging error. The angular error between the gyro output axis and the output reference axis when the gyro is erected/caged shall be within the limits specified in the detail specification.

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* 3.6.3.7.5 Minimum nutation frequency. The nutation frequency (see 6.3.17) shall not fall below a specified minimum value.

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* 3.6.3.7.6 Maximum free drift rate. The free drift rate of the vertical gyroscope with the spin axis erection system deactivated shall not exceed the specified maximum value.

* 3.6.3.7.7 Maximum systematic drift rate (with specified conditions). The systematic drift rate (see 6.3.18) of the vertical gyroscope with the spin axis erection system deactivated, and with the gyroscope subjected to specified dynamic environment conditions shall not exceed the specified maximum value.

* 3.6.3.7.8 Maximum random drift rate. The random drift rate of the gyro when subjected to the specified dynamic environment shall not exceed the specified minimum levels.

* 3.6.3.7.9 Acceleration-sensitive drift rate. The drift rate caused by accelerations acting on the mass unbalance along any principal axis or gimbal shall not exceed the specified maximum level.

* 3.6.3.8 Polarity. The polarity of the positive electrical output signal of the pick-offs and torquers is given as follows:

* 3.6.3.8.1 Pick-offs. The polarity of each pick-off shall be such that the component of the spin axis angular displacement along the indicated output axis is a positive electrical signal when the missile has undergone a positive angular displacement in attitude about the corresponding pitch or roll missile axis (see figure 1).

* 3.6.3.8.2 Torquers. The polarity of each torquer shall be such that with a positive command input, precessional displacement of the spin axis will occur about the orthogonal axis corresponding to positive pitch and roll.

3.6.4 Type IV - directional gyroscope. These requirements shall apply to the type IV gyroscope only.

* 3.6.4.1 Reference axis identification. The two orthogonal gyroscope output axes shall be the "pitch" and yaw axes. The spin reference axis (SRA) shall be roll axis. The gyroscope orientation

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with respect to the missile is fixed by its type (directional) designation. The gyroscope yaw axis shall correspond to the missile yaw axis, but the gyroscope "pitch" axis need not correspond to the missile pitch axis if a means is provided for rotating the yaw axis pick-off reference such that it is at nominal zero position for the rotor spin axis lying in any yaw orientation within the plane defined by the missile pitch and roll reference axes.

* 3.6.4.2 Gimbal freedom. The maximum angular displacement capability of both gimbals shall be as specified in the detail specification. (see 6.3.3) The gimbal freedom need not be equal in both axes.

* 3.6.4.3 Torquer. When a "pitch" axis torquer is required for rotor spin axis precession about the yaw axis it shall be so specified by the procuring activity (see 6.2), and torquer number 1 requirements shall apply.

* 3.6.4.3.1 Minimum torquing rate. The torquer shall be capable of applying a continuous torque such that the precession rate about the yaw axis orthogonal to the torquer axis is at least a specified minimum rate. The torquer may be rated instead on an intermittent duty basis for higher torques, as suitable for the application. When so specified, the torquer shall be capable of satisfying both requirements.

3.6.4.4 Spin axis leveling system. The directional gyroscope shall incorporate a means for maintaining the rotor spin axis in a nominally horizontal position, with a leveling reference as specified below.

* 3.6.4.4.1 Polarity. The polarity of the pick-off outputs shall be coincident with the corresponding missile angular displacement polarity.

* 3.6.4.4.2 Leveling reference. The detail specification shall establish whether the leveling reference shall be the horizontal plane defined by the pitch and roll principal reference axes of the missile (the "case-leveled" directional gyroscope) or a perpendicular to the local vertical defined by the gravity field of the Earth (the "local horizontal-referenced" directional gyroscope).

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* 3.6.4.4.2.1 Leveling sensor operational range. The operational ranges of the leveling sensor used for gravity-perpendicular leveled directional gyroscopes shall be such that the proper signal polarity (and amplitude, if proportional leveling is used) shall be maintained throughout the missile attitude ranges specified in the detail specification.

* 3.6.4.4.3 Excitation. The nominal RMS voltage, frequency, and continuous power available for excitation of the reference sensor shall be specified in the detail specification.

* 3.6.4.4.4 Leveling rate. The nominal leveling rate(s) at rotor design speed and the maximum plus or minus percentage variation of this leveling rate from the nominal value shall be specified in the detail specification.

* 3.6.4.4.5 Leveling accuracy. The directional gyroscope shall satisfy the minimum leveling accuracy as specified in the detail specification for both leveling axes.

* 3.6.4.5 Gyro scale factor. The gyro scale factor (see 6.3.10) shall conform to the detail specification.

3.6.4.6 Performance. The directional gyroscope shall conform to the following performance requirements as specified in the detail specification.

* 3.6.4.6.1 Maximum composite error. The composite error (see 6.3.16) of the gyroscope applicable to the yaw output axis shall not exceed the specified maximum value.

* 3.6.4.6.2 Maximum electrical zero (at angular displacement nominal zero). The excited pick-off output voltage when the rotor spin axis is horizontal and aligned with the spin reference axis (SRA) or roll principal reference axis (or the pick-off reference is rotated to its nominal "null" heading position regardless of the azimuth or yaw direction of the spin axis), shall not exceed a specified value.

* 3.6.4.6.3 Minimum nutation frequency. The nutation frequency (see 6.3.17) shall not fall below a specified minimum value.

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* 3.6.4.6.4 Maximum free drift rate. The free drift rate with the spin axis leveling system deactivated shall not exceed the specified maximum value.

* 3.6.4.6.5 Maximum systematic drift rate (with specified conditions). The systematic drift rate (see 6.3.18) of the directional gyroscope with the spin axis leveling system deactivated, and with the gyroscope subjected to specified dynamic environment conditions, shall not exceed a specified maximum value.

* 3.6.4.6.6 Maximum random drift rate. The random drift rate of the gyroscope, when subjected to specified dynamic environments, shall not exceed the specified maximum value.

* 3.6.4.6.7 Acceleration-sensitive drift rate. The drift rate caused by accelerations acting upon the mass unbalance along any principal axis or gimbal shall not exceed the specified maximum level of the detail specification.

3.6.4.7 Polarity. The polarity of the positive electrical output signal of the pick-off and the positive electrical input command signal of the torquer are given as follows.

3.6.4.7.1 Pick-off. The polarity of the yaw axis pick-off shall be such that it provides a positive electrical signal when the missile undergoes a positive angular displacement in attitude about the missile yaw axis (see figure 1).

* 3.6.4.7.2 Torquer. The polarity of the torquer shall be such that when a positive command input electrical signal is applied, it produces an angular precession of the spin axis about the orthogonal yaw axis in a positive direction.

3.6.5 Type V - rate integration gyroscope. These requirements shall apply to the type V gyroscope only.

* 3.6.5.1 Reference axis identification. The gyroscope output axis or OA shall be identified on the detail system drawings as the pitch, yaw, or roll axis in accordance with its relation to the principal missile reference axes of figure 1. The gyroscope spin reference axis (SRA) and input reference axis (IRA) shall be identified on the detail system drawings as the pitch, yaw, or roll axes in similar fashion.

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* 3.6.5.2 Input ranges. The maximum input range (see 6.3.15.5) shall be as specified in the detail specification.

* 3.6.5.3 Mechanical overload. For a specified time duration, the rate integrating gyroscope shall be capable of withstanding an angular rotation rate about the input reference axis (IRA) of either polarity and of a maximum magnitude specified in the detail specification. It shall suffer no damage or subsequent impairment of performance.

3.6.5.4 Pick-off. Unless otherwise specified (see 6.2), the type II pick-off shall be the only type used for the rate integrating gyroscope.

* 3.6.5.5 Torquer. A linear torquer shall be required for the output axis. The type II torquer or any other (type III) torquer which is a nominally linear transducer of input signal to output torque shall be the only torquer type used for the rate integrating gyroscope.

3.6.5.5.1 Minimum range of equivalent input rate. A type II torquer shall be capable of applying a continuous range of torques of each polarity about the output axis such that the torque produced by the maximum gyroscope angular rotation rate about the input reference axis (IRA) can be equalled or exceeded by the corresponding counter-torque within the torquer range of linear output. The maximum gyroscope angular rotation rates which correspond to the limits of these output torque ranges shall be specified for each polarity.

* 3.6.5.6 Gyro scale factor. The scale factor (see 6.3.10) shall conform to the detail specification.

3.6.5.7 Performance. The rate integration gyroscope shall conform to the following performance requirements as specified in the detail specification.

3.6.5.7.1 Characteristic time. The approximate characteristic time and appropriate tolerance limits for its variation under the full range of specified operational environment conditions shall be specified in the detail specification.

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* 3.6.5.7.2 Maximum composite error. The composite error (see 6.3.16) of the rate integrating gyroscope shall not exceed a specified maximum value. Unless otherwise specified (see 6.2), the rate integrating gyroscope shall also conform separately to the requirements of the following components of the composite error:

a. Maximum threshold

b. Maximum resolution

3.6.5.7.3 Maximum electrical null (at angular displacement nominal zero). The excited pick-off output voltage waveform when the rotor axis is aligned with the spin reference axis (SRA), shall not exceed a specified RMS value.

* 3.6.5.7.4 Maximum random drift rate. The maximum random drift rate shall be as specified in the detail specification.

* 3.6.5.7.5 Maximum systematic drift rate. The maximum systematic drift rate (see 6.3.18) shall be as specified in the detail specification.

* 3.6.5.7.6 Maximum acceleration-sensitive drift rate. The maximum acceleration-sensitive drift rate shall be specified in terms of the equivalent input rate of angular rotation of the gyroscope about the input reference axis (IRA).

* 3.6.5.7.7 Maximum accleration squared-sensitive drift rate. The maximum acceleration squared-sensitive drift rate shall be specified in terms of the equivalent input rate of angular rotation of the gyroscope about the input reference axis (IRA).

3.6.5.7.8 Angular sensitivity about the spin reference axis (SRA). Unless otherwise specified by the procuring activity (see 6.2), the maximum angular sensitivity about the spin reference axis (SRA) under various conditions of gyroscope angular displacement about the input reference axis (IRA), shall be specified as follows.

3.6.5.7.8.1 At zero angular displacement about the input reference axis (IRA). The rate integrating gyroscope shall have less than a specified maximum sensitivity to angular rotation about the spin reference axis (SRA) with zero angular displacement of the gyroscope about the input reference axis (IRA).

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3.6.5.7.8.2 At specified angular displacement about the input reference axis (IRA). The rate integrating gyroscope shall have less than a specified maximum sensitivity to angular rotation about the spin reference axis when a specified angular displacement is applied about the input reference axis (IRA). The angular displacement of the gyroscope about the input reference axis (IRA) shall be at least one half the maximum input range.

3.6.5.8 Polarity. The polarity of the positive electrical output signal of the pick-off and the positive electrical input command signal of the torquer are given as follows.

* 3.6.5.8.1 Pick-off. The polarity of the pick-off shall be such that its indicated angular rotation of the spin axis about the output axis shall be a positive electrical signal as defined in the basic diagram of the detail specification.

* 3.6.5.8.2 Torquer. The polarity of the torquer shall be such that when a positive command input electrical signal is applied, it shall produce a positive torque as defined in the basic diagram of the detail specification.

3.7 Environmental effects. Unless otherwise specified (see 6.2), the gyroscope shall be capable of withstanding the effects of the environmental conditions stated in sub-paragraphs 3.7.1, 3.7.2, and 3.7.4 (for non-hermetically sealed gyroscopes only) without impairment of its specified performance.

3.7.1 Storage and transit. Unless otherwise specified (see 6.2), the gyroscope shall be designed and packaged for shipment in such a way that there shall be no impairment of its required functional capabilities following exposure of the packaged gyroscope to the climatic extremes specified for short term storage and transit in MIL-STD-210, and following exposure to the shock test of Procedure III, Method 516.1 of MIL-STD-810.

* 3.7.2 Operational environment. Unless otherwise required in the detail specification, the gyroscope shall be capable of withstanding the environments specified herein when it is installed in the missile system.

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3.7.2.1 Thermal equilibrium. The gyroscope shall exhibit no impairment of its specified performance when subjected to the environments contained in the test Methods 501.1 and 502.1 of MIL-STD-810. The maximum and minimum equilibrium ambient temperatures at which the gyroscope must operate shall be specified.

3.7.2.2 Thermal shock. The gyroscope shall exhibit no impairment of its specified performance when subjected to the environments contained in the test Method 503.1 of MIL-STD-810.

3.7.2.3 Low pressure. The gyroscope shall exhibit no impairment of its specified performance when subjected to the environments contained in the test Method 500.1 (procedure II) of MIL-STD-810. The temperature shall be the low temperature limit specified in 3.7.2.1 of this specification, instead of the value given. The minimum pressure at which the gyroscope must be operated and the rate of pressure change (as required test information) shall be specified.

3.7.2.4 Temperature-altitude (pressure) cycling. The gyroscope shall exhibit no impairment of its specified performance when subjected to the environments contained in the test Method 504.1 of MIL-STD-810; with the modifications that in steps 2 and 3 the temperature shall be lowest temperature at which the gyroscope must operate, specified in 3.7.2.1 of this specification, and in steps 6 through 13, the maximum temperature allowable during any period of gyroscope operation shall be the highest temperature at which the gyroscope must operate, specified in 3.7.2.1 of this specification. The equipment class designation (as required test information) shall be specified. The equipment operational mode shall be considered as continuous.

3.7.2.5 External magnetic fields. When specified (see 6.2), the gyroscope shall be capable of specified performance during exposure to the effects of specified levels of magnetic flux density in the respective principal reference axis direction.

* 3.7.2.6 Radiation. The gyroscope shall exhibit no impairment of its specified performance under normal environment conditions, following exposure to nuclear radiation of a type, intensity, and duration specified. MIL-STD-446 or other acceptable methods (see 6.2) shall be used to determine applicable exposure and test procedures.

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3.7.2.7 Vibration. The gyroscope shall exhibit no impairment of its specified performance when subject to the environments contained in the test Method 514.1 of MIL-STD-810. The equipment class (as required test information) pertaining to the gyroscope, and the equipment mounting method shall be specified.

* 3.7.2.8 Shock. The gyroscope shall exhibit no impairment of its specified performance when subjected to the environments contained in the test Method 516.1 of MIL-STD-810. Whether the gyroscope is to be operating or non-operating during the shock test, whether the gyroscope is to be caged or uncaged (when applicable) during either of these conditions, and whether Procedure I, Procedure IV, or Procedure V is to be used, shall be specified. When specified (see 6.2), distinctly different multiple shock tests for applicable combinations of these conditions shall be used to more accurately simulate actual shock environments for certain high shock gyroscope applications.

* 3.7.2.9 Acceleration. When this requirement is specified (see 6.2), the gyroscope shall exhibit no impairment of its specified performance when subjected to the environments contained in the test Method 513.1 of MIL-STD-810. Procedure II shall be used, and the test method shall be by rectilinear acceleration as in a rocket sled test, rather than by centripetal acceleration as in a centrifuge test.

* 3.7.3 Hermetic seal. When hermetic seal is specified (see 6.2), the gyroscope shall be hermetically sealed in accordance with the definition of MIL-STD-108 unless otherwise specified in the detail specification. When applicable, this requirement may apply to the gyroscope as it exists up to the instant of initiation of operation, and shall not apply during operation of the gyroscope. The hermetic seal requirements shall be considered as fulfilled when the maximum leak rate measured in the applicable test of Method 112 of MIL-STD-202 is less than a maximum allowable leak rate specified in the detail specification.

* 3.7.4 Operational environment (non-hermetic gyroscope). Unless otherwise specified in the detail specification, the environmental requirements contained in this paragraph are required only for gyroscopes which are not hermetically sealed throughout their storage, transit, and operational lifetime.

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3.7.4.1 Humidity. The gyroscope shall exhibit no impairment of its specified performance when subjected to the environments contained in the test Method 507.1 of MIL-STD-810. Procedure I shall be used.

* 3.7.4.2 Fungus. The gyroscope shall exhibit no impairment of its specified performance when subjected to the environments contained in the test Method 508.1 of MIL-STD-810. Either Procedure I or II may be used, but if Procedure II is used it shall be applied to all materials that can support fungus growth as used in the structure of the gyroscope, including bonding agents, coatings, encapsulation, and dielectric materials.

3.7.4.3 Salt spray. The gyroscope shall exhibit no impairment of its specified performance when subjected to the environments contained in the test Method 509.1 of MIL-STD-810. Procedure I shall be used.

3.7.4.4 Sand and dust. The gyroscope shall exhibit no impairment of its specified performance when subjected to the environments contained in the test Method 510.1 of MIL-STD-810. Procedure I shall be used.

* 3.8 Interchangeability. Certain corresponding assemblies, components, and parts of a given gyroscope of the same model and series, shall be functionally and physically interchangeable.

* 3.9 Maximum weight. The gyroscope shall be designed to yield its specified performance with the overall weight of the unit to be the least amount practicable and consistent with minimum manufacturing cost. A maximum, and minimum if applicable, allowable total unit weight shall be specified.

* 3.10 Maximum exterior dimensions. The maximum, and minimum if applicable, exterior physical dimensions of the gyroscope (in a three dimensional orthogonal coordinate axis sense) shall be specified.

* 3.11 Electromagnetic interference. Unless otherwise specified (see 6.2), the gyroscope shall meet the electromagnetic interference requirements of MIL-STD-461 for the applicable equipment class, but no test report shall be required.

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3.12 Identification of product. The gyroscope shall be marked for identification in accordance with MIL-STD-130. The gyroscope type as determined from 1.2 of this specification, and named in the detail specification, shall be a part of the nomenclature as required in MIL-STD-130.

* 3.12.1 Axis marking. If the gyroscope does not have a sufficiency of distinguishing superficial features to readily and uniquely define the gyroscope principal reference axes (e.g. a gyroscope in a cylindrical case), it shall have visible axis markings on the gyroscope case exterior, with only one marking required for each axis. These axis markings which are entirely separate from the identification marking (see 3.12) shall be affixed to the gyroscope case intercept of the gyroscope principal reference axes as named below for each of the gyroscope types:

Type I	-	Spin Reference Axis (SRA)
Type II	-	Input Reference Axis (IRA) Spin Reference Axis (SRA)
Type III	-	Spin Reference Axis (SRA)
Type IV	-	Spin Reference Axis (SRA)
Type V	-	Input Reference Axis (IRA) Spin Reference Axis (SRA)

The positive rotation polarity sense shall be in accordance with the polarities shown for rotations defined in the basic diagram of the detail specification.

* 3.13 Workmanship. The gyroscope and all of its parts shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, welding and brazing, surface treatments, painting, impregnation or potting of coils, machine screw assemblies, marking of parts and assemblies, and the freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications. Where practicable, Requirement 9 of MIL-STD-454 shall apply in addition to the workmanship requirements stated herein.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of examinations and tests. The inspection and testing of any gyroscopes of the types defined in 1.2 shall be classified as follows:

a. Preproduction tests

b. Acceptance examinations and tests

c. Periodic production tests and sampling

4.3 Test conditions. Unless otherwise specified by the procuring activity (see 6.2) or otherwise specified in the test method description, the following general test conditions shall apply to all tests of this specification.

* 4.3.1 Standard atmospheric conditions. Whenever the gyroscope pressure and temperature existing at the time of the test are not definitely specified, it is understood that the test will be made at atmospheric pressure (28.5 + 2.0 or -4.5 inches Hg) and at room temperature (23 $\pm$ 10 deg. C.). When tests are made with the pressure or temperature differing materially from the above values, proper allowance shall be made for the difference from the specified conditions.

* 4.3.2 Test equipment accuracy. Accuracy of the test equipment selected to measure the specified equipment performance requirements shall be based on the applicable National Bureau of Standards and equipment design limits. The test equipment shall be selected for technical compatibility with the equipment during test to assure specified accuracy of test readout.

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4.3.3 Electric power input conditions. Unless otherwise specified in the particular test method, the gyroscope shall be tested with the nominal excitation voltages, at the nominal frequencies, applied at the appropriate inputs.

4.3.4 Output load impedance. Unless otherwise specified in the particular test method, all pick-off outputs of the gyroscope shall be connected to an electrical load simulating the input impedance and loading effect of the equipment to be connected to these pick-off outputs during service. When practicable, the actual equipment may be used.

4.4 Preproduction tests. Unless otherwise specified (see 6.2), gyroscopes shall be subjected to preproduction tests as set forth herein. The preproduction tests of the gyroscope shall consist of all tests of this specification as specified under 4.7 as applicable to the gyroscope type.

* 4.4.1 Preproduction test samples. The preproduction test samples shall consist of a number of gyroscopes all of the same manufacturer, and manufactured using the same methods and procedures proposed for future production. The number of preproduction test samples submitted shall be specified in the contract or purchase order (see 6.2). Approval of the preproduction sample by the procuring activity shall be based on the requirements and tests applicable to the specific gyroscope type. Further production of the gyroscope by the supplier, prior to the approval of the preproduction sample, shall be at the risk of the supplier.

* 4.4.2 Preproduction test report. When the tests are conducted at a location other than the laboratory of the procuring activity, the following shall be furnished to that activity upon completion of the tests:

a. Test report - three copies of a test report prepared in accordance with MIL-STD-831

b. The samples which were tested

c. Two sets of assembly or installation drawings, as applicable

d. Brief operating instructions

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4.5 Acceptance examinations and tests. Acceptance examinations and tests shall be performed on all gyroscopes manufactured and submitted for acceptance under contract. Unless otherwise specified (see 6.2), each gyroscope delivered under contract shall be subjected to all of the examinations and tests of this paragraph applicable to the gyroscope type. Failure to meet any of the requirements associated with these tests shall be considered cause for rejection.

* 4.5.1 Type I - free gyroscope. Each free gyroscope shall be subjected to the following tests as detailed in the applicable test method subparagraph of 4.7.

a. Examination of product (4.7.1)

b. Insulation resistance test (4.7.2)

c. Leak test (hermetically sealed gyroscopes only) (4.7.3)

d. Angular transfer characteristic test (free gyroscope) (4.7.4)

e. Free drift test (free gyroscope) (4.7.5)

f. Torque to balance drift rate test (free gyroscope) (4.7.6)

* 4.5.2 Type II - rate gyroscope. Each rate gyroscope shall be subjected to the following tests as detailed in the applicable test method subparagraph of 4.7.

a. Examination of product (4.7.1)

b. Insulation resistance test (4.7.2)

c. Leak Test (hermetically sealed gyroscopes only) (4.7.3)

d. Rate transfer characteristic test (4.7.7)

e. Tumble test (rate gyroscope) (4.7.8)

*

* 4.5.3 Type III - vertical gyroscope. Each vertical gyroscope shall be subjected to the following tests as detailed in the applicable test method subparagraph 4.7.

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a. Examination of product (4.7.1)

b. Insulation resistance test (4.7.2)

c. Leak test (hermetically sealed gyroscopes only) (4.7.3)

d. Angular transfer characteristic test (vertical gyroscope) (4.7.10)

e. Free drift rate test (vertical gyroscopes) (4.7.11)

* 4.5.4 Type IV - directional gyroscope. Each directional gyroscope shall be subjected to the following tests as detailed in the applicable test method subparagraph of 4.7.

a. Examination of product (4.7.1)

b. Insulation resistance test (4.7.2)

c. Leak test (hermetically sealed gyroscopes only) (4.7.3)

d. Angular transfer characteristic test (directional gyroscope) (4.7.12)

e. Free drift rate test (directional gyroscope) (4.7.13)

* 4.5.5 Type V - rate integrating gyroscope. Each rate integrating gyroscope shall be subjected to the following tests as detailed in the applicable test method subparagraph 4.7.

a. Examination of product (4.7.1)

b. Insulation resistance test (4.7.2)

c. Leak test (hermetically sealed gyroscopes only) (4.7.3)

d. Angular transfer characteristic test (rate integrating gyroscope) (4.7.14)

e. Random drift rate test (rate integrating gyroscopes) (4.7.15)

f. Tumble test (rate integrating gyroscopes) (4.7.16)

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4.6 Periodic production tests and sampling plan. Periodic production tests are those tests which are performed on a sample of the gyroscopes manufactured and submitted for acceptance under contract. Production samples shall be randomly selected (on a regular basis determined by production rate) from gyroscopes that have satisfactorily passed the acceptance examinations and tests of 4.5. Unless otherwise specified (see 6.2), the sample gyroscopes shall be subjected to all the tests listed under this paragraph that are applicable to the gyroscope type. The sampling plan to be used and number of samples to be selected will be in accordance with MIL-STD-105.

4.6.1 Sampling plan tests (all gyroscope types). Each sample gyroscope of any type shall be subjected to the following tests as applicable to the gyroscope with respect to the fact that it may be hermetically sealed or not hermetically sealed, and as detailed in the applicable test method subparagraph of 4.7.

* 4.6.2 Type I - free gyroscope. Each sample free gyroscope shall be subjected to the following tests as detailed in the applicable test method subparagraph of 4.7.

a. Electromagnetic interference test (4.7.17)

b. Explosive atmosphere test (4.7.18)

c. Torquer test (free gyroscope)(4.7.19)

d. Dynamic drift test (free gyroscope)(4.7.20)

* 4.6.3 Type II - rate gyroscope. Each sample rate gyroscope shall be subjected to the following tests as detailed in the applicable test method subparagraph of 4.7.

a. Electromagnetic interference test (4.7.17)

b. Explosive atmosphere test (4.7.18)

c. Mechanical overload test (rate gyroscope)(4.7.21)

d. Torquer test (rate gyroscope)(4.7.22)

e. Dynamic response test (rate gyroscope)(4.7.23)

f. Maximum rate sensitivity about the spin reference axis test (at zero angular rate about the input reference axis)(4.7.24)

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g. Maximum rate sensitivity about the spin reference axis test (at specified angular rate about the input reference axis) (4.7.25)

h. Maximum rate sensitivity about the output axis test (4.7.26)

i. Maximum angular acceleration sensitivity about the output axis test (4.7.27)

* 4.6.4 Type III - vertical gyroscope. Each sample vertical gyroscope shall be subjected to the following tests as detailed in the applicable test method subparagraph of 4.7.

a. Electromagnetic interference test (4.7.17)

b. Explosive atmosphere test (4.7.18)

c. Maximum erection error test (vertical gyroscope) (4.7.28)

d. Dynamic drift test (vertical gyroscope) (4.7.29)

* 4.6.5 Type IV - directional gyroscope. Each sample directional gyroscope shall be subjected to the following tests as detailed in the applicable test method subparagraph of 4.7.

a. Electromagnetic interference test (4.7.17)

b. Explosive atmosphere test (4.7.18)

c. Torquer test (directional gyroscope) (4.7.30)

d. Dynamic drift test (directional gyroscope) (4.7.31)

* 4.6.6 Type V - rate integrating gyroscope. Each sample rate integrating gyroscope shall be subjected to the following tests as detailed in the applicable test method subparagraph of 4.7.

a. Electromagnetic interference test (4.7.17)

b. Explosive atmosphere test (4.7.18)

c. Mechanical overload test (rate integrating gyroscope) (4.7.32)

d. Dynamic response test (rate integrating gyroscope) (4.7.33)

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e. Maximum angular sensitivity about spin reference axis (at zero angular displacement about the input reference axis) (4.7.34)

f. Maximum angular sensitivity about spin reference axis (at specified angular displacement about the input reference axis) (4.7.35)

4.7 Test methods.

4.7.1 Examination of product. The gyroscope shall be inspected to determine compliance with the requirements specified herein with respect to materials, finish, marking, and workmanship in general, and with particular attention to the following:

a. Identification of product (see 3.12)

b. Axis marking (see 3.12.1)

4.7.2 Insulation resistance test. The gyroscope shall be subjected to the test procedure detailed in Method 302 of MIL-STD-202. The minimum direct potential to be applied for each barrier test shall be specified in the detail specification.

* 4.7.3 Leak test (hermetically sealed gyroscopes only). The selection of the appropriate leak test shall be determined by the internal configuration of the hermetic gyroscope; one type of test shall be specified for the outer case of the gas-filled, non-floated gimbal gyroscope, while another shall be specified for the outer case of the fluid-filled, floated gimbal gyroscope.

4.7.3.1 Leak test (non-floated gimbal). The gyroscope shall be subjected to the test procedure detailed in Method 112 of MIL-STD-202. Test condition C and Procedure IV shall be used. Perforation shall not be required and bubble test condition B shall be used to test for gross leaks. The reduced pressure of the mass spectrometer leak test chamber, the duration of the evacuation exposure, and the leakage-rate sensitivity of the mass spectrometer test shall be specified in the detail specification.

4.7.3.2 Leak test (floated gimbal). The gyroscope shall be placed in a pressure chamber (dry air may be used as the gas) and subjected to a gauge pressure to two atmospheres for a minimum period of time specified in the detail specification. It shall

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be removed from the chamber and visually examined under low power magnification in good light for evidence of fluid leakage at any point along the length of the outer case seal. Any sign of fluid leakage shall be cause for rejection of the gyroscope.

* 4.7.4 Angular transfer characteristic test (free gyroscope). This test shall measure the composite error of the pick-off indicated rotor spin axis position with respect to rotation about the output axes, the cross-coupling effect, and the pick-off nominal zero displacement angle of the free gyroscope. It shall also directly determine the observed maximum electrical null (at angular displacement nominal zero), and the observed maximum signal null (at angular displacement nominal zero). From this data, the observed maximum composite error (both axes), the observed maximum cross-coupling error can be determined, in addition to those parameters determined directly.

* 4.7.4.1 Test equipment. A two axis displacement table shall be required. The test surface to which the gyroscope is mounted shall be capable of sufficient angular rotation about its perpendicular axes that the free gyroscope can be positioned in any attitude throughout the full ranges of its output axes. A means of demodulating the pick-off signal (if demodulation is required) shall be provided, and a means of accurately determining the displacement angles of the gyroscope mounting surface about the angular freedom axes of the table shall be required.

* 4.7.4.2 Test setup and procedure. Mount the gyroscope on the two axis displacement table with the gyroscope output axes coincident with the table angular freedom axes, and the spin reference axis (SRA) of the gyroscope perpendicular to the plane of the table axes when the table is in its initial reference position. The table axes correspond to the appropriate missile principal axes of figure 1.

With the table in the initial reference position, and the gyroscope caged, operate the gyroscope and the pick-offs, and record the observed pick-off maximum electrical null in RMS volts, for both pick-offs, measured at the pick-off outputs before the demodulators, Record the pick-off maximum signal null in RMS volts, for both pick-offs, measured at the pick-off output or any point suitable for isolating the signal component of the pick-off output voltage waveform.

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Uncage the gyroscope and as rapidly as possible (commensurate with the necessities of data taking or plotting) displace the table about only one table axis in the positive angular displacement direction of the corresponding missile principal axis until the maximum displacement angle is reached which corresponds to the limit of the positive angular freedom range of the gyroscope. Now displace the table in the negative angular displacement direction through the initial reference position and continue until the maximum displacement angle is reached which corresponds to the limit of the negative angular freedom range of the gyroscope. Finally displace the table in the positive angular displacement direction once again until the table initial reference position is reached, and stop on that point. Rapid performance of this test cycle is essential to minimize the effects of gyroscope drift upon this measurement. Recage the gyroscope and repeat this test cycle for the other table axis.

* 4.7.4.3 Test results. Draw the specified output function line on the data plots. This line is a plot of the "ideal" pick-off (demodulated) signal output vs. the rotor spin axis angular displacement about the output axis; that is, a plot of the pick-off design scale-factor. The maximum observed deviation of the output from this line divided by the sum of the output ranges (output span) expressed as a percentage, is the maximum composite error of that output axis. The larger of the two percentage values computed for each of the plots shall be considered the observed maximum composite error of the gyroscope.

Divide the maximum magnitude observed value of the plot of the "other" pick-off output (the pick-off for the other output axis about which there is nominally no angular displacement) by the sum of the output ranges (output span) of the pick-off which indicates the angular displacement about the table axis of rotation, and express as a percent. Do this for both "other" pick-off plots, and the larger of the two percentage values computed for each of the plots shall be considered the observed maximum cross-coupling error.

Locate the input axis crossing of each data plot (for the output axis about which there is angular displacement) and mark the input axis point midway between these crossings. If hysteresis is small there may be only one input axis crossing on each plot. The input axis distance between this midpoint and the origin of the plot shall be expressed in degrees and called the reference displacement

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error of that output axis. The larger of the two values found (one for each of the plots) shall be considered the observed maximum reference displacement error.

* 4.7.5 Free drift test (free gyroscope). This test shall measure the free drift as a function of time for a free gyroscope without torquers, with the gyroscope in a stationary attitude with respect to the Earth-fixed coordinates. From a plot of this free drift vs. time, the maximum observed free drift, and the maximum observed free drift rate, may be determined. This test is intended primarily for lower performance free gyroscopes for which the specified maximum free drift rate is much larger than the Earth's rate, and which generally do not have torquers specified as a requirement.

4.7.5.1 Test equipment. A single axis displacement table shall be required. The test surface to which the gyroscope is mounted shall be capable of at least 90 degree rotation about the table axis so that the gyroscope can be positioned with its spin reference axis (SRA) aligned with the local vertical (gravity determined) and any perpendicular local horizontal. A means of demodulating the pick-off signal (if demodulation is required) shall be provided, and a means of accurately determining the alignment of the spin reference axis (SRA) to the local vertical or horizontal (via some convenient table mounting surface reference axis), shall be required.

4.7.5.2 Test setup and procedure. Mount the gyroscope on the single axis displacement table with the spin reference axis (SRA) coincident with the gravity-determined local vertical. The single axis table is locked in position. Operate the gyroscope and pick-offs, and when equilibrium rotor speed is reached, record the (demodulated) signal output of each pick-off with the gyroscope caged. Then uncage the gyroscope and begin drift data recording for the (demodulated) output of both pick-offs as a function of time. Allow the data recording to continue for a "reasonable" length of time, such as a typical missile flight time duration, or the specified operation time of a drive and coast gyroscope. Shut down the gyroscope at the end of the data taking interval in such a way that there is no risk of damage to the gyroscope.

Repeat this test with the gyroscope spin reference axis (SRA) aligned to any local horizontal.

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4.7.5.3 Test results. Determine the maximum deviation (occurring at any time within the test interval) of the (demodulated) signal output of each pick-off from the output of each when the gyroscope is caged. When converted to an angle using the pick-off design scale factor, the largest of the four values obtained (one value for each pick-off, two tests) shall be considered the observed maximum free drift.

* The maximum observed value of the free drift rate for a given pick-off in a single test, shall be determined by finding the maximum value of the quotient of a drift angle value divided by the time elapsed from the instant of uncaging to the time of occurrence of that drift value. The results obtained are then adjusted by the vertical component of Earth's rate for the test latitude. To avoid difficulty in calculation and unrealistic results, the calculation of the quotient shall not be performed during the time interval from the instant of uncaging to 10% of the elapsed time of the test. The calculation of the quotient may be performed continuously in analog fashion so that the peak value of the function may be found more readily, making this method amenable to automatic test procedures. The largest of the four values obtained (one value for each pick-off, two tests) shall be considered the observed maximum free drift rate.

* 4.7.6 Torque to balance drift rate test (free gyroscope). This test shall measure the free drift rate as a function of time for a free gyroscope which has two type II torquers, or any other (type III) two torquers which are nominally linear transducers of input signal to output torque. From a plot of torquer signal vs. time, with the torquer design scale factor known, the maximum observed drift rate may be determined. This test is intended primarily for high performance gyroscopes of the continuous drive type which are provided with proportional (type II) torquers on both output axes.

4.7.6.1 Test equipment. A single axis displacement table shall be required. The test surface to which the gyroscope is mounted shall be capable of at least 90 degree rotation about the table axis so that the gyroscope can be positioned with its spin reference axis (SRA) aligned with the local vertical (gravity determined) and any perpendicular local horizontal. A means of demodulating the pick-off signal (if demodulation is required) shall be provided and a means of accurately determining the alignment of the spin reference axis (SRA) to the local vertical or horizontal (via some convenient table mounting surface reference axis), shall be

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required. A means of demodulating (if required) and amplifying the pick-off signals to be applied to each of the orthogonal torquer inputs so as to always drive the spin axis to the spin reference axis position in the fashion of the "capture loop" technique, shall be required.

4.7.6.2 Test setup and procedures. Mount the gyroscope on the single axis displacement table with the spin reference axis (SRA) coincident with the gravity-determined local vertical. The single axis table is locked in position. Operate the gyroscope, pick-offs, and torquers. When equilibrium rotor speed is reached, record the torquer input signal (demodulated if not available in analog form) of each torquer with the gyroscope caged. Then uncage the gyroscope and begin torquer input signal recording of both torquers as a function of time. Allow the data recording to continue for a "reasonable" length of time, such as a typical missile flight time duration. Shut down the gyroscope at the end of the data taking interval in such a way that there is no risk of damage to the gyroscope.

Repeat this test with the gyroscope spin reference axis (SRA) aligned to any local horizontal.

* 4.7.6.3 Test results. Determine the maximum deviation (occurring at any time within the test interval) of the (demodulated, if necessary) input signal of each torquer when the gyroscope is caged. When converted to an angular rate using the torquer design scale factor (signal to torque) and the calculated gyroscope precession scale factor (torque about an output axis to rate of angular rotation of the spin axis about the other output axis), the largest of the four values obtained (one value for each pick-off, two tests) shall be considered the observed maximum drift rate. The maximum drift is not directly available from this data without resorting to integration techniques.

* 4.7.7 Rate transfer characteristic test. This test shall measure the composite error of the pick-off indicated rotor spin axis positions with respect to rotation about the output axis and the pick-off nominal zero displacement angle of the rate gyroscope. It shall also directly determine the observed maximum electrical null (at angular displacement nominal zero), and the observed maximum signal null (at angular displacement nominal zero). From this data, the observed maximum composite error and the observed displacement error can be determined, in addition to those parameters determined directly.

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4.7.7.1 Test equipment. A single axis rate table shall be required. It shall be capable of calibrated angular rotation rates of sufficient magnitude in each direction (CW or CCW) to reach the angular rotation rate extremes of the input ranges of the rate gyroscope under test. A means of accurately determining the table rate (if not an integral part of that equipment) as well as a means of demodulating the pick-off signal (if demodulation is required) shall be provided.

4.7.7.2 Test setup and procedure. Mount the rate gyroscope with the input reference axis (IRA) parallel to the rate table axis of rotation, which in turn is nominally aligned with the (gravity determined) local vertical. With the rate table stopped (zero angular rate input) operate the gyroscope and pick-off, and record the pick-off maximum electrical null in RMS volts, measured at the pick-off output before the demodulator. Record the pick-off maximum signal null in RMS volts, measured at the pick-off output or any point suitable for isolating the signal component of the pick-off output waveform. Arrange to take data or plot the (demodulated) output of the pick-off as a function of table angular rate.

Start the table at a low angular rate in one direction, and increase the table angular rate to the angular rotation rate extreme of the corresponding input range of the rate gyroscope. This is preferably done as a continuous gradual increase of table rate, or slowly in small increments, to avoid any transient dynamic response effects of the gyroscope, the rate table, or the plotting apparatus. Decrease the table angular rate to the zero angular input rate and then continue through zero to increase the table angular rate in the opposite direction to the angular rotation rate extreme of the corresponding (other) input range of the rate gyroscope. Finally, decrease the table angular rate to the zero angular input rate once again to complete one cycle of the input span, in one direction sense.

Repeat this cycle of the input span in the opposite direction sense; that is, increase the table angular rate in the last "opposite" direction first, and continue through the entire test cycle, to be sure that any effects of nonrepeatability are manifest.

* 4.7.7.3 Test results. Draw the specified output function line on the data plots. The maximum observed deviation of the output from this line (for either direction sense, full cycle of the input span) divided by the sum of the output ranges (output span) expressed as percentage, is the observed maximum composite error of the gyroscope.

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Locate the input axis crossings of each data plot (for each direction sense, full cycle of the input span) and mark the input axis point midway between these crossings. If hysteresis is small there may be only one input axis crossing on each plot, and if non-repeatability is negligible the two plots may be identical. The input axis distance between this midpoint and the origin of the plot shall be expressed in degrees and called the reference displacement error of the output axis. The largest of the two values found shall be considered the observed maximum reference displacement error.

In general, the accuracy requirements of the rate gyroscopes under test are sufficiently low so that the effect of the Earth's rate can be neglected, but with higher accuracy rate gyroscopes, correction for the Earth's rate may be desirable. Other test procedures more suitable to this higher level of rate gyroscope precision should also be considered.

* 4.7.8 Tumble test (rate gyroscope). This test shall measure the drift of the pick-off indicated rotor spin axis position from the spin reference axis (SRA), with respect to gyroscope rotation about the output axis caused by a component of the Earth's gravity field. From a plot of this drift vs. gyroscope principal reference axis system attitude and with the reference displacement error taken into account, the maximum observed acceleration-sensitive drift rate may be determined.

4.7.8.1 Test equipment. A single axis displacement table shall be required. The test surface to which the rate gyroscope is mounted shall be capable of 360 degree or continuous rotation about the table axis. As an alternative a single axis rate table may be used. A means of demodulating the pick-off signal (if demodulation is required) shall be provided, and a means of accurately determining the displacement angle of the gyroscope mounting surface about the table angular freedom axis shall be provided.

4.7.8.2 Test setup and procedure. Position the single axis displacement table so that the table axis is aligned with any local horizontal (perpendicular to the gravity-determined local vertical). Mount the rate gyroscope with the output axis (gyroscope principal reference axis) parallel to the table axis, or stated in a different way, with the input reference axis (IRA) and the spin reference axis (SRA) mutually perpendicular to the table axis. With the

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table stationary, operate the gyroscope and pick-off, and then start rotation of the displacement table or the rate table about its table axis in either direction. The rate of table angular rotation about its table axis shall be smooth and slow as to avoid any transient dynamic response effects of the rate gyroscope or the plotting apparatus. The rate or table angular rotation about its axis may be constant, but this is not essential so long as the dynamic response effects are negligible. Continue the table angular rotation about its table axis, in the chosen direction, until at least one complete revolution is attained.

* 4.7.8.3 Test results. Determine the maximum positive (demodulated) output deviation of the pick-off occurring at any table displacement position within the chosen interval of table rotation. Determine the maximum negative (demodulated) output deviation of the pick-off occurring at any table angular displacement position within the chosen interval of table rotation. Add the two magnitudes (taken as absolute values without regard to algebraic sign), then subtract the observed maximum (magnitude) signal null (at angular displacement nominal zero) which is determined in the rate transfer characteristic test of 4.7.7. Divide the result by two, then convert to the equivalent angular input rate using the rate gyroscope scale factor. The final result expressed as the equivalent angular input rate per unit acceleration in g's, shall be considered the observed maximum acceleration-sensitive drift rate.

* 4.7.9 Vibration test (rate gyroscope). This test shall measure the drift rate of the pick-off indicated rotor spin axis position from the spin reference axis position (SRA), with respect to angular rotation about the output axis, as a function of a vibrational acceleration frequency and certain RMS acceleration magnitudes, applied along a single axis line bisecting the 90 degree angle formed by the input reference axis (IRA) and the spin reference axis (SRA). From a plot of this drift rate vs. vibrational acceleration frequency and certain RMS acceleration magnitudes, the frequencies of resonance and their associated large drift rates can be measured. The largest of these drift rate magnitudes, with the reference displacement error and the acceleration-sensitive drift rate components taken into account, allows determination of the maximum observed acceleration squared-sensitive drift rate. This test is not intended to be an environmental vibration test,

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as the levels of vibration used should be sufficiently high to accomplish the measurements, but not as high as those levels used in severe vibration environment tests.

4.7.9.1 Test equipment. A single axis vibration machine shall be required, as well as a test fixture for mounting the rate gyroscope in a stationary orientation so that the single axis line which bisects the 90 degree angle formed by the input reference axis (IRA) and the spin reference axis (SRA) is coincident with the axis of vibration of the test surface of the machine. A means of demodulating the pick-off signal (if demodulation is required) and means of accurately determining the frequency and the RMS amplitude of the periodic (sinusoidal) acceleration shall be provided. If the vibration machine does not have a constant frequency independent vibration acceleration output, or if it does not have automatic gain compensation to maintain such an output, it may be necessary to calibrate the vibration machine for the particular test gyroscope and fixture mass utilized in this test. This calibration may be performed with a "dummy" load of the same mass and shape as the test gyroscope and fixture. The functional relationship of the machine driving source gain to frequency required to obtain constant RMS vibration acceleration output at each of the selected vibration levels, can then be experimentally determined.

4.7.9.2 Test setup and procedure. Mount the rate gyroscope and its mounting fixture such that the single axis line bisects the 90 degree angle formed by the input reference axis (IRA) and the spin reference axis (SRA) is coincident with the test surface vibration axis. With the vibration machine turned off, operate the gyroscope and pick-off, and record the (demodulated) signal output of the pick-off. Start the vibration machine, and at each specified vibration acceleration amplitude level adjust the vibration frequency to slowly traverse the specified range of frequencies. It is necessary to traverse the frequency range slowly so that the recorder dynamic response does not affect the results, and so that sharp resonant peaks, if they exist, are determined accurately in amplitude.

* 4.7.9.3 Test results. From each of the plots of (demodulated) signal output of the pick-off vs. frequency, select the largest amplitude of signal output occurring at any frequency. Subtract the signal output of the pick-off recorded with the vibration machine turned off, then divide this result by the square of the RMS vibration acceleration level at which the plot was made. Select the largest of these results and convert it to the equivalent angular

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input rate (per acceleration squared) using the rate gyroscope specified scale factor. The final result expressed as the equivalent angular input rate per unit acceleration squared in g's, shall be considered the observed maximum acceleration squared-sensitive drift rate.

* 4.7.10 Angular transfer characteristic test (vertical gyroscope). This test shall measure the composite error of the pick-off indicated rotor spin axis position with respect to rotation about the output axes, the cross-coupling effect, and the pick-off nominal zero displacement angle of the vertical gyroscope. It shall also directly determine the observed maximum electrical null (at angular displacement nominal zero), and the observed maximum signal null (at angular displacement nominal zero). From this data, the observed maximum composite error (both axes), the observed maximum cross-coupling error, and the observed reference displacement error can be determined, in addition to those parameters determined directly. It is not the purpose of this test to separate out the effect of erection error. Rather, the effect of erection error is implicit in all of these parameters (except for cross-coupling error); both those directly determined and those calculated from the measurements.

* 4.7.10.1 Test equipment. A two axis displacement table shall be required. The test surface to which the gyroscope is mounted shall be capable of sufficient angular rotation about its perpendicular axes of freedom that the vertical gyroscope can be positioned in any attitude throughout the full ranges of its output reference axes (pitch and roll axes). A means of demodulating the pick-off signal (if demodulation is required) shall be provided, and a means of accurately determining the displacement angles of the gyroscope mounting surface about the angular freedom axes of the table shall be required.

* 4.7.10.2 Test setup and procedure. Position the two axis displacement table so that the angular freedom axes of the table are each mutually perpendicular to the gravity-determined local vertical. Mount the vertical gyroscope on the table with the gyroscope output reference axis (pitch and roll axes) coincident with the table angular freedom axes. The spin reference axis (SRA) or yaw axis of the gyroscope shall be coincident with the table angular freedom axes. The spin reference axis (SRA) or yaw axis of the gyroscope shall be coincident with the gravity-determined local

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vertical when the table is in its initial reference position. The table axes correspond to the pitch and roll principal reference axes of the missile.

With the table in the initial reference position, operate the gyroscope and the pick-offs. After allowing a sufficient interval of time for the vertical gyroscope spin axis to erect to its final equilibrium position, record the maximum electrical null in RMS volts, for both pick-offs, measured before the demodulators. Also record the maximum signal null in RMS volts, for both pick-offs, measured at their outputs or any points suitable for isolating the signal component of the output voltage waveform. Plot the (demodulated) output of both pick-offs as a function of table displacement angle about one table axis.

* As rapidly as possible (commensurate with the limitations of the dynamic response effects of the recording apparatus), displace the table about only one table axis in the positive angular displacement direction of the corresponding missile principal axis until the maximum displacement angle is reached which corresponds to the limit of the positive input range of the gyroscope. Now displace the table in the negative angular displacement direction through the initial reference position and continue until the maximum displacement angle is reached which corresponds to the limit of the negative input range of the gyroscope. Finally displace the table in the positive angular displacement direction once again until the table initial reference position is reached, and stop on that point. This test cycle should be performed rapidly to conserve test time, but not so rapidly as to introduce dynamic response effects of the recording apparatus into the data. The rapidity of the test has no significant effect on the observed performance of the vertical gyroscope under these test conditions.

Repeat this test cycle for the other table axis.

* 4.7.10.3 Test results. Draw the specified output function line on the data plots. This line is a plot of the "ideal" pick-off (demodulated) signal output vs. the rotor spin axis angular displacement about the output axis; that is, a plot of the pick-off design scale factor. The maximum observed deviation of the output from this line divided by the sum of the output ranges (output span) expressed as a percentage, is the maximum composite error of that output axis. The larger of the two percentage values computed for each of the plots shall be considered the observed maximum composite error of the gyroscope.

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Divide the maximum magnitude observed value of the plot of the "other" pick-off output (the pick-off for the other output axis about which there is nominally no angular displacement) by the sum of the output ranges (output span) of the pick-off which indicates the angular displacement about the table axis of rotation, and express as a percent. Do this for both "other" pick-off plots, and the larger of the two percentage values computed for each of the plots shall be considered the observed maximum cross-coupling error.

Locate the input axis crossings of each data plot (for the output axis about which there is angular displacement) and mark the input axis point midway between these crossings. If hysteresis is small, there may be only one input axis crossing on each plot. The input axis distance between this midpoint and the origin of the plot shall be expressed in degrees and called the reference displacement error of that output axis. The largest of the two values found for each of the plots shall be considered the observed maximum reference displacement error.

4.7.11 Free drift rate test (vertical gyroscope). This test shall measure, as a function of time, the rotor spin axis free angular drift from the initial erected position (nominally aligned with the spin reference axis and the gravity determined local vertical, or a case determined local vertical) for a vertical gyroscope. This shall be measured with the spin axis erection system deactivated, and with the gyroscope in a stationary attitude with respect to the Earth-fixed coordinates. From the plot of this free drift vs. time, the maximum observed free drift, and the maximum observed free drift rate, may be determined.

4.7.11.1 Test equipment. A means of demodulating the pick-off signal (if demodulation is required) shall be provided, and a means of accurately determining the alignment of the spin reference axis (SRA) to the local vertical (via some convenient table mounting surface reference axis) shall be required.

* 4.7.11.2 Test setup and procedure. Mount the gyroscope on a suitable table or fixture so that the spin reference axis (SRA) is coincident with the gravity-determined local vertical. Operate the gyroscope and the pick-offs, but not the torquers or any of the associated spin axis erection system equipment. When the vertical gyroscope has operated for a sufficient length of time such that the

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design rotor speed has been reached, uncage the gyroscope, and begin drift data recording for the (demodulated) output of both pick-offs as a function of time. If caging is not provided for the vertical gyroscope under test, start the gyroscope in the usual way with the spin axis erection system active, and after the gyroscope has reached its specified performance condition (the rotor rotation at design speed, and the spin axis fully erected), deactivate the spin axis erection system (for both output axes), and begin drift data recording. Allow the data recording to continue for a "reasonable" length of time, such as a typical missile flight time duration. Shut down the gyroscope at the end of the data taking interval in such a way that there is no risk of damage to the gyroscope.

* 4.7.11.3 Test results. Determine the maximum deviation (occurring at any time within the test interval) of the (demodulated) output of each pick-off from the output of each when the gyroscope is caged, or in its initial condition with the spin axis fully erected. When converted to an angle using the pick-off design scale factor, the larger of the two values obtained (one value for each pick-off) shall be considered the observed maximum free drift.

The maximum observed value of the free drift rate for a given pick-off shall be determined by finding the maximum value of the quotient of a drift angle value divided by the time elapsed from the instant of uncaging (or the beginning of the drift data recording) to the time of occurrence of that drift value. To avoid difficulty in calculation and unrealistic results, the calculation of the quotient shall not be performed during the time interval from the instant of uncaging to 10% of the elapsed time of the test. The calculation of the quotient may be performed continuously in analog fashion so that the peak value of the function may be found more readily, making this method amenable to automatic test procedures.

NOTE: From the calculated value of the total drift rate obtained in these tests, algebraically subtract the effect of the vertical component of Earth's rate.

* 4.7.12 Angular transfer characteristic test (directional gyroscope). This test shall measure the composite error of the pick-off indicated rotor spin axis position with respect to rotation about the output axis (yaw axis), the cross-coupling effect (only for the effect of attitude changes about the "pitch" output axis) and the observed maximum electrical null (at the signal null angular position). From this data, the observed maximum composite error, and the observed maximum

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cross-coupling error can be determined, in addition to the parameter determined directly. It is not the purpose of this test to separate out the effect of leveling error. Rather, the effect of leveling error is implicit in all of these parameters (except for cross-coupling error); both those directly determined and those that are calculated from the measurements.

* 4.7.12.1 Test equipment. A two axis displacement table shall be required. The test surface to which the gyroscope is mounted shall be capable of sufficient angular rotation about its perpendicular axes of freedom that the directional gyroscope under test can be positioned in any attitude throughout the full ranges of its output axes (yaw and "pitch" axes). A means of demodulating the pick-off signal (if demodulation is required) shall be provided, and a means of accurately determining the displacement angles of the gyroscope mounting surface about the angular freedom axis of the table shall be required.

4.7.12.2 Test setup and procedure. Position the two axis displacement table so that the axis of largest angular freedom is coincident with the gravity-determined local vertical. The other perpendicular table axis is aligned with any local horizontal. Mount the directional gyroscope on the two axis displacement table so that the gyroscope yaw axis (principal reference axis) is aligned with the table axis coincident with the local vertical, and the gyroscope spin reference axis (SRA) is perpendicular to the table axis which is aligned to a local horizontal. The gyroscope "pitch" axis is now aligned with any arbitrary local horizontal. Under these conditions, and including subsequent corrections, the table axes will correspond to the yaw and pitch principal reference axes of the missile as shown in figure 1.

With the table in the initial reference position with regard to the horizontal table axis, operate the gyroscope and the pick-offs. After allowing a sufficient interval of time for the directional gyroscope spin axis to level to some local horizontal equilibrium position, monitor the pick-off (demodulated) signal output and rotate the two axis table about the angular freedom axis coincident with the local vertical until the pick-off signal output reaches zero. The rotor spin axis is then nominally coincident with the spin reference axis (SRA), and the gyroscope "pitch" axis is nominally aligned with the table horizontal axis of freedom. The directional gyroscope is now in its initial reference position. This procedure

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unavoidably removes the reference displacement error, so that the rotor spin axis alignment with the spin reference axis (SRA) and the gyroscope "pitch" axis alignment with the table horizontal axis is strictly "nominal", since there is usually no independent means (other than the pick-off) of ascertaining the rotor spin axis alignment error unless the directional gyroscope has caging or the mechanism is accessible to optical measurement.

Now record the maximum electrical null in RMS volts, measured at the pick-off output before the demodulator. Plot the (demodulated) output of the pick-off as a function of table displacement angle about the table axis coincident with the local vertical. As rapidly as possible (commensurate with the limitations of the dynamic response effects of the recording apparatus), displace the table about the table axis coincident with the local vertical in the positive angular displacement direction of the corresponding missile principal axis until the maximum displacement angle is reached which corresponds to the limit of the positive angular freedom range of the gyroscope's yaw axis. Now displace the table in the negative angular displacement direction through the initial reference position and continue until the maximum displacement angle is reached which corresponds to the limit of the negative angular freedom range of the gyroscope's yaw axis. Finally displace the table in the positive angular displacement direction once again until the table initial reference position is reached, and stop on that point. If the directional gyroscope has a 360 degree (continuous) angular freedom span about the yaw axis, then the two ranges may be chosen as 180 degrees each. This test cycle should be performed rapidly to conserve test time, but not so rapidly as to introduce dynamic response effects of the recording apparatus into the data. The rapidity of the test has no significant effect on the observed performance of the directional gyroscope under these test conditions.

Repeat this test cycle for the horizontal table axis.

* 4.7.12.3 Test results. Draw the specified output function line on the first data plot. This line is a plot of the "ideal" pick-off (demodulated) signal output vs. the rotor spin axis angular displacement about the output axis; that is, a plot of the pick-off design scale factor. The maximum observed deviation of the output from this line divided by the sum of the output range (output span) expressed as a percentage, is the observed maximum composite error of the output axis (yaw axis).

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Divide the maximum observed value of the second plot of the pick-off output (the plot obtained when the test was repeated for the horizontal table axis) by the sum of the output ranges (output span) of the pick-off as obtained in the first plot, and express as a percent. This shall be considered the observed maximum cross-coupling error.

* 4.7.13 Free drift rate test (directional gyroscope). This test shall measure, as a function of time, the rotor spin axis free angular drift rate from the initial leveled position (nominally aligned with a local horizontal perpendicular to a gravity determined local vertical, or a case determined local horizontal) about the gyroscope yaw axis for a directional gyroscope. This shall be measured with the spin axis leveling system deactivated, and with the gyroscope in a stationary attitude with respect to the Earth-fixed coordinates. From the plot of this free drift vs. time, the maximum observed free drift, and the maximum observed free drift rate, may be determined.

* 4.7.13.1 Test equipment. A single axis displacement table shall be required which is capable of 360 degree rotation of the gyroscope mounting surface about the table axis of freedom. A means of demodulating the pick-off signal (if demodulation is required) shall be provided.

* 4.7.13.2 Test setup and procedure. Position the single axis displacement table so that the angular freedom axis of the table is aligned with the gravity-determined local vertical. Mount the directional gyroscope on the single axis displacement table so that the gyroscope yaw output axis is aligned with the table angular freedom axis. With the spin axis leveling system active, operate the gyroscope and the pick-offs. After allowing a sufficient interval of time for the directional gyroscope spin axis to level to some local horizontal equilibrium position, monitor the pick-off (demodulated) signal output and rotate the table about its angular freedom axis until the pick-off signal output reaches zero. The directional gyroscope rotor spin axis is then nominally coincident with the spin reference axis (SRA). Now deactivate the spin axis leveling system and begin drift data recording. Allow the data recording to to continue for a "reasonable" length of time, such as a typical missile flight time duration. Shut down the gyroscope at the end of the data taking interval in such a way that there is no risk of damage to the gyroscope.

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* 4.7.13.3 Test results. Determine the maximum deviation (occurring at any time within the test interval) of the (demodulated) output of each pick-off from the zero signal output condition set at the start of the drift data recording interval. When converted to an angle using the pick-off design scale factor, the value obtained shall be considered the observed maximum free drift. The test results obtained must then be adjusted to remove Earth's rate from the calculated drift rate values.

The maximum observed value of the free drift rate shall be determined by finding the maximum value of the quotient of a drift angle value divided by the time elapsed from the beginning of the drift data recording to the time of occurrence of that drift value. To avoid difficulty in calculation and unrealistic results, the calculation of the quotient shall not be performed during the time interval from the beginning of the drift data recording to 10% of the elapsed time of the test. The calculation of the quotient may be performed continuously in analog fashion so that the peak value of the function may be found more readily, making this method amenable to automatic test procedures.

* 4.7.14 Angular transfer characteristic test (rate integrating gyroscope). This test shall measure the composite error of the pick-off indicated rotor spin axis position with respect to rotation about the output axis. It shall also directly determine the observed maximum electrical null (at the signal null angular position). From this data the observed maximum composite error can be determined, in addition to that parameter determined directly.

4.7.14.1 Test equipment. A single axis displacement table shall be required. The test surface to which the gyroscope is mounted shall be capable of sufficient angular rotation about its table axis of freedom that the rate integrating gyroscope can be positioned at any displacement angle about its input reference axis (IRA), throughout the full input span. A means of demodulating the pick-off signal (if demodulation is required) shall be provided as well as a means of accurately determining the displacement angle of the gyroscope mounting surface about the table angular freedom axis. Equipment for amplifying the pick-off (demodulated) signal output to be applied to the torquer input so that the rotor spin axis may be driven to the spin reference axis position shall also be required.

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* 4.7.14.2 Test setup and procedure. Mount the rate integrating gyroscope on the single axis displacement table with the gyroscope input reference axis (IRA) parallel with the table angular freedom axis. The table axis corresponds to an appropriate missile principal reference axis of figure 1. Plot the (demodulated) signal output of the pick-off as a function of displacement angle about the table angular freedom axis. With the table in the initial reference position, and with the arrangements made to amplify the pick-off output signal and apply it to the torquer input in the proper polarity so as to drive the pick-off output signal to zero, operate the gyroscope. Allow a sufficient interval of operation so that equilibrium is reached with the pick-off output signal driven to zero and the gyroscope rotor spin axis nominally aligned with the spin reference axis (SRA). Record the maximum electrical null in RMS volts, measured at the pick-off output before the demodulator. Then remove the signal from the torquer input, and as rapidly as possible (commensurate with the necessities of data taking or plotting) displace the table about its axis in the positive angular displacement direction of the corresponding missile axis until the maximum displacement angle is reached which corresponds to the limit of the gyroscope input range. Then displace the table in the negative angular displacement direction through the initial reference position and continue until the maximum displacement angle is reached which corresponds to the limit of the gyroscope negative input range. Finally, displace the table in the positive angular displacement direction once again until the table initial reference position is reached, and stop on that point. Rapid performance of this test cycle is essential to minimize the effects of gyroscope drift upon this measurement.

4.7.14.3 Test results. Draw the specified output function line on the data plot. The maximum observed deviation of the output from this line (for the full cycle of the input span) divided by the sum of the output ranges (output span) expressed as a percentage, is the observed maximum composite error of the gyroscope.

* 4.7.15 Random drift rate test (rate integrating gyroscope). This test shall measure the instantaneous drift rate of a rate integrating gyroscope as a function of time, with the gyroscope in a specific fixed orientation with respect to the Earth-fixed coordinates. From a plot of this instantaneous drift rate vs. time, the maximum observed random drift rate may be determined.

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4.7.15.1 Test equipment. A means of demodulating the pick-off signal (if demodulation is required) shall be provided, as well as a means of accurately determining the alignment of the rate integrating gyroscope output axis to the Earth's axis of rotation (via some convenient table mounting surface reference axis). Equipment for amplifying the pick-off (demodulated) output signal to be applied to the torquer input so that the rotor spin axis may be driven to the spin reference axis position shall also be required.

* 4.7.15.2 Test setup and procedure. Mount the gyroscope on the test surface so that the gyroscope output axis is parallel with the Earth's axis of rotation. With arrangements made to amplify the pick-off (demodulated) output signal and apply it to the torquer input in the proper polarity so as to drive the pick-off output signal to near zero, operate the gyroscope. Allow a sufficient interval of operation so that dynamic equilibrium is reached with the pick-off output signal driven to near zero and the gyroscope rotor spin axis nominally aligned with the spin reference axis (SRA). Begin torquer input signal recording as a function of time. Allow the data recording to continue for a "reasonable" length of time, such as a typical missile flight time duration. Shut down the gyroscope at the end of the data taking interval in such a way that there is no risk of damage to the gyroscope.

* 4.7.15.3 Test results. The average value of the torquer signal over the entire data recording interval shall be determined by convenient graphical or analytic techniques. A new function of torquer signal vs. time shall be calculated for the entire data recording interval by subtracting this average torquer signal from all previously recorded instantaneous torquer signal values. This new function has a zero average value. Calculate the RMS value based on the entire data recording interval for this new function of torquer signal vs. time. When this value is converted to an equivalent drift rate using the torquer measured scale factor (signal to torque) and the calculated gyroscope precession scale factor (torque about the output axis to angular rotation rate of the spin axis about the input reference axis), the result shall be the RMS value of the measured instantaneous drift rate with all real and apparent systematic drift rate removed. This shall be considered the maximum observed random drift rate.

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* 4.7.16 Tumble test (rate integrating gyroscope). This test shall measure the instantaneous drift rate of a rate integrating gyroscope as a function of the gyroscope attitude with respect to the Earth-fixed coordinates, while the rate integrating gyroscope is mounted on a sidereal rate table. From a record of the maximum observed Earth's rate of this instantaneous drift rate vs. sidereal table angle, the maximum observed systematic acceleration-insensitive drift rate, the maximum observed systematic acceleration-sensitive drift rate, and the maximum observed systematic acceleration squared-sensitive drift rate, may be determined in one hour, 11 minutes, and 48.2 seconds test time.

* 4.7.16.1 Test equipment. A means of rotating the gyroscope about a single axis parallel with the Earth's axis of rotation at 20 times Earth's rate of one revolution every one hour, 11 minutes, and 48.2 seconds shall be provided, usually with a sidereal rate table. A means of demodulating the pick-off signal (if demodulation is required) shall be provided; as well as a means of accurately determining the alignment of the rate integrating gyroscope output axis to the Earth's axis of rotation (via some convenient table mounting surface reference axis). Also required, shall be a means of accurately determining the sidereal table angular position with respect to an Earth-fixed index mark and equipment for amplifying the pick-off (demodulated) output signal for application to the torquer input so that the rotor spin axis may be driven to the spin reference axis position.

* 4.7.16.2 Test setup and procedure. The sidereal rate table must be positioned so that the table axis of rotation is parallel to the Earth's axis of rotation. Mount the gyroscope on the test surface of the table so that the gyroscope output axis is aligned with the table axis of rotation. With arrangements made to amplify the pick-off (demodulated) output signal and apply it to the torquer input in the proper polarity so as to drive the pick-off output signal to near zero, start the sidereal table and operate the gyroscope. Allow a sufficient interval of operation so that dynamic equilibrium is reached with the pick-off output signal driven to near zero and the gyroscope rotor spin axis nominally aligned with the spin reference axis (SRA). At a particular recorded instant of time and table position, begin the torquer input signal recording as a function of time (or sidereal table position). Allow the data recording to continue for one complete revolution of the table. Shut down the gyroscope and the sidereal table at the end of the data taking interval in such a way that there is no risk of damage to the gyroscope.

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* 4.7.16.3 Test results. The record of the torquer input signal over the entire test interval corresponds to one revolution of the sidereal table with respect to the Earth-fixed coordinates, and as a consequence, the local gravity vector (gravity-determined local vertical) has made one complete revolution with respect to the gyroscope. With the exclusion of random drift rate effect, and the insensitive drift rate, the systematic drift rate effects which are acceleration (gravity) dependent can be expected to repeat if the data recording were extended for another full sidereal table revolution. Thus these effects are periodic with sidereal table rotation, the period being one table revolution. These periodic effects can be obtained from the data by a method of Fourier analysis which takes advantage of this periodicity to isolate these periodic components and the constant component, but ignores the aperiodic (random) component.

When the Fourier coefficients for the common representation of the truncated series shown have been determined (or the corresponding coefficients of the terms of any other preferred representation),

$$f(x) \approx A_o + a_1 \cos x + b_1 \sin x + a_2 \cos 2x + b_2 \sin 2x$$

the components of the torquer signal may be calculated from the expressions following:

a. Average component proportional to systematic acceleration-insensitive drift rate = A_o

b. Fundamental component proportional to systematic acceleration-sensitive drift rate

$$= \sqrt{a_1^2 + b_1^2}$$

c. Second harmonic component proportional to systematic acceleration squared-sensitive drift rate

$$= \sqrt{a_2^2 + b_2^2}$$

The average component A_o is converted to an equivalent drift rate using the torquer measured scale factor (signal to torque) and the calculated gyroscope precession scale factor (torque about the output axis or angular freedom reference axis to angular rotation rate

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of the spin axis about the input reference axis). The result shall be the maximum observed acceleration-insensitive drift rate.

The fundamental component $\sqrt{a_1^2 + b_1^2}$ is converted to an equivalent drift rate using the torquer measured scale factor and the calculated gyroscope precession scale factor. The result is divided by the cosine of the latitude angle of the test site (since only the component of gravitational acceleration normal to the output axis is effective in producing a drift rate), so that the final result expressed as the equivalent angular input drift rate per unit acceleration in g's, shall be considered the observed maximum acceleration-sensitive drift rate.

The second harmonic component $\sqrt{a_2^2 + b_2^2}$ is converted to an equivalent drift rate using the torquer measured scale factor and the calculated gyroscope precession scale factor. The result is divided by the cosine of the latitude angle of the test site so that the final result, expressed as the equivalent angular input drift rate per unit acceleration squared in g's squared, shall be considered the observed maximum acceleration squared-sensitive drift rate.

* 4.7.17 Electromagnetic interference test. The test methods to be used for this test shall be in accordance with MIL-STD-462 for the applicable equipment class as specified in MIL-STD-461.

4.7.18 Explosive atmosphere test. The gyroscope shall be subjected to the test procedure detailed in Procedure I of Method 511.1 of MIL-STD-810. The type of fuel or volatile combustible material used in the test to form the explosive atmosphere shall be specified in the detail specification. Procedure II of Method 511.1 of MIL-STD-810 shall not apply in this test.

4.7.19 Torquer test (free gyroscope). This test shall measure the composite error of the type II (proportional) torquer output torque with respect to the input signal D.C. voltage, and with nominally zero rotor spin axis displacement from the spin reference axis (SRA). From this data the observed torquer maximum composite error can be determined.

4.7.19.1 Test equipment. A single axis servo-driven turntable shall be required, with a means provided for accurately determining the turntable angular rotation rate about the axis of rotation.

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A means of demodulating (if demodulation is required) and amplifying the pick-off signal to be applied to the turntable servo amplifier input shall be provided. This signal shall be applied in the proper polarity so as to drive the pick-off (demodulated) signal output to zero (and maintain the rotor spin axis in nominal alignment with the spin reference axis). Equipment for generating a torquer input signal DC voltage capable of ranging from zero to the maximum input signal DC voltage in either polarity, shall also be required.

* 4.7.19.2 Test setup and procedure. Mount the gyroscope on the single axis servo turntable so that the gyroscope output axis orthogonal to the output axis on which the torquer is located is coincident with the servo turntable axis of rotation. The servo turntable axis and the coincident gyroscope axis correspond to an appropriate missile principal reference axis. Connect the output of the pick-off located on this gyroscope axis to the input of the turntable servo amplifier in the proper polarity to rotate the turntable in the direction which drives this pick-off (demodulated) signal to zero. Connect torquer input signal source to the input of the torquer located on the orthogonal gyroscope axis.

With the gyroscope caged, operate the gyroscope, and when equilibrium operating conditions have been reached, uncage the gyroscope. As rapidly as possible (commensurate with the response dynamics of the turntable servo control system, and the necessities of data taking or plotting) increase, from initial zero, the torquer input signal DC voltage in the polarity direction causing the turntable to rotate in the positive angular displacement direction of the corresponding missile principal axis. Continue to increase the torquer input signal in this direction until the maximum torquer input signal and the corresponding maximum rate of gyroscope spin axis precession (and identical turntable rotation rate) about the servo turntable axis is reached. Now decrease the torquer input signal through zero (and through the corresponding turntable zero rotation rate) and continue to the opposite voltage polarity until the identical condition is reached in the negative angular displacement direction. Finally, increase the torquer input signal once again until zero torquer input signal and the corresponding zero rotation rate are reached, and stop on that point. Rapid performance of this test cycle is essential to minimize the effects of gyroscope drift upon this measurement. Shut down the gyroscope at the end of this test cycle in such a way that there is no risk of damage to the gyroscope. Recage the gyroscope.

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With the proper reorientation of the gyroscope on the servo turntable and reconnection of the turntable servo amplifier to the other pick-off, repeat this test for the other torquer, if applicable.

* 4.7.19.3 Test results. Draw the torquer specified output function line on the data plots. This line is drawn on each plot by converting the torquer output torque variable of the torquer specified output function to an equivalent gyroscope angular precession rate using the calculated gyroscope precession scale factor (torque about an output axis to angular rotation rate of the spin axis about the orthogonal output axis). This provides a function of the "ideal" gyroscope angular precession rate vs. the torquer input signal DC voltage. The maximum observed deviation of the precession rate output from this line divided by the sum of the precession rate ranges (span) expressed as a percentage, is the maximum composite error for that torquer. The larger of the two percentage values computed for each of the plots shall be considered the observed torquer maximum composite error.

4.7.20 Dynamic drift test (free gyroscope). This test shall measure the rotor spin axis free angular drift from the initial caged angular position (nominally aligned with the spin reference axis), for a free gyroscope with or without torquers. This shall be measured with the gyroscope subjected to an intervening interval of the dynamic operating conditions imposed by a two axis sinusoidal motion (Scorsby) table.

4.7.20.1 Test equipment. A two axis sinusoidal motion (Scorsby) table shall be required. A gyroscope mounting fixture which is capable of at least a 90 degree rotation shall be required so that the gyroscope can be positioned with its spin reference axis (SRA) approximately aligned with the local vertical or any local horizontal perpendicular to the local vertical. A means of demodulating the pick-off signal (if demodulation is required) shall be provided.

4.7.20.2 Test setup and procedure. Mount the gyroscope on the table-mounted fixture, and align the spin reference axis (SRA) to an approximate local vertical while the table is stopped in the reference (leveled) position (displaced from any instantaneous position in its normal displacement cycle). The table oscillation frequency shall be specified in the detail specification. The amplitude of the table displacement angles shall be such that the rotor

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spin axis is displaced through at least 50% of the angular freedom ranges. Operate the gyroscope, and when equilibrium rotor speed is reached, record the (demodulated) signal output of each pick-off with the gyroscope caged. Uncage the gyroscope, then tilt the table so that it is in the correct attitude phase position of the continuous two axis sinusoidal motion cycle. Start the table oscillating and continue for a "reasonable" length of time, such as a typical missile flight time duration, or the specified operation time of a drive and coast gyroscope. In any case, the time interval shall be the same as that used in the free drift test of 4.7.5.

At the end of this test interval stop the table motion but allow the gyroscope to continue to run while the table is quickly returned to the reference position. Record the signal output of each pick-off, and shut down the gyroscope in such a way that there is no risk of damage to the gyroscope.

Repeat the test with the gyroscope spin reference axis (SRA) approximately aligned to any local horizontal with the table in the reference position.

4.7.20.3 Test results. Determine the deviation of the (demodulated) signal output of each pick-off as the recorded output at the end of the test interval minus the recorded output before gyroscope uncaging. When converted to an angle using the pick-off design scale factor, the largest of the four values obtained (one value for each pick-off, two tests) shall be considered the observed maximum free drift under dynamic conditions. This value shall be within the limit for this parameter as specified without regard to the dynamic oscillation effect.

* 4.7.21 Mechanical overload test (rate gyroscope). The sample rate gyroscope shall be operated while an overload input rotation is applied about the input reference axis. The test equipment required and the setup shall be similar to that of the rate transfer characteristic test (see 4.7.7). No monitoring of rate gyroscope performance is required during the application of the input overload rates. Apply the specified input overload rate first in one direction sense for the specified time interval (see 3.6.2.3), and then likewise in the other direction sense. Finally, subject this sample rate gyroscope to the rate transfer characteristic test of 4.7.7. If the gyroscope then provides satisfactory performance under the conditions of this test, it shall be considered to have fulfilled all performance requirements associated with the input overload test.

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4.7.22 Torquer test (rate gyroscope). This test shall measure the performance of the rate gyroscope (self-test) torquer, and from this measurement data, it shall determine the minimum range of equivalent input rate for each input polarity. This test is not intended to measure the composite error, or any of the other errors associated with the torquer, since its use in the rate gyroscope is mainly for self-test or preflight checkout purposes, and not for systematic rate drift error correction.

4.7.22.1 Test equipment. A means of demodulating the pick-off signal (if demodulation is required), as well as a means of generating and measuring a torquer signal of either polarity to be applied to the torquer input, shall be required.

4.7.22.2 Test setup and procedure. With the rate gyroscope mounted in any convenient position, operate the gyroscope and the pick-off with zero input signal applied to the torquer. Calculate the pick-off (demodulated) signal outputs corresponding to the respective limits of the input ranges (see 3.6.2.2) using the gyroscope specified output function (see 3.6.2.5). Slowly increase the torquer input signal in one polarity sense so that the monitored pick-off signal begins to increase correspondingly. Continue to increase the torquer input signal until reaching the calculated pick-off signal corresponding to the limit of the input range. Aside from verifying that this torquer signal input level can be reached, record the torquer signal input level as evidence that this level was successfully attained.

Repeat this procedure for the other torquer input signal polarity and the other corresponding input range of the gyroscope.

4.7.22.3 Test results. The levels of torquer input signal required to produce a pick-off (demodulated) signal output for the limits of respective input ranges shall be within the limits imposed by the maximum DC voltage available for torquer excitation (see 3.5.6.3.2 and 3.5.6.4.2).

4.7.23 Dynamic response test (rate gyroscope). This test shall measure the dynamic response of the rate gyroscope to a variable frequency sinusoidal input rate about the input reference axis. From this data, the approximate undamped natural frequency and the approximate damping ratio may be determined. The validity and utility of these parameters are limited in their application to

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the actual rate gyroscope by the closeness of the approximation of its dynamic response to the well known and tabulated dynamic response of the ideal "second order system" (see 6.3.12).

4.7.23.1 Test equipment. An oscillating rate table shall be required, with means provided for adjustment of the frequency and amplitude of the sinusoidal rate over an adequate range. The frequency range of the oscillating rate table shall extend to a limit well above the expected nominal undamped natural frequency of the rate gyroscope design, and the angular rate amplitude shall be within the input range of the gyroscope.

Also required, are a means of accurately determining the frequency and phase of the table input angular rate (if not an integral part of that equipment), a means of demodulating the pick-off signal (if demodulation is required) and a means of determining the phase difference between the pick-off sinusoidal output signal (or the fundamental component of the demodulated output signal waveform) and the table sinusoidal input rate.

4.7.23.2 Test setup and procedure. Mount the rate gyroscope on the test surface of the oscillating rate table with the input reference axis (IRA) parallel to the table axis. Plot the phase difference of the pick-off (demodulated) signal output and the table input rate as a function of table input frequency. With the table stopped, operate the rate gyroscope and pick-off. Start the table in oscillation at a low frequency and slowly increase the frequency of the table rate input while noting the phase difference which is being recorded. Continue to increase the table input frequency until the phase of the gyroscope pick-off signal lags the phase of the table input rate by 90 degrees. Record the frequency at which this occurs, and continue to increase the frequency above this value until either the limit of the frequency range of the oscillating rate table is reached, or the amplitude of the oscillation has decreased to a level where reliable phase readings of the pick-off signal and the table input can no longer be obtained. Stop the oscillating rate table and shut down the gyroscope. The slow rate of increase in the frequency of oscillation is to avoid any transient dynamic effects of the rate table or the plotting apparatus, so that the true steady state dynamic response is obtained.

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4.7.23.3 Test results. The frequency at which the phase of the pick-off (demodulated) signal output lags the phase of the table input rate by 90 degrees shall be considered as the observed undamped natural frequency. The plot of the phase difference as a function of table rate input frequency is normalized by dividing each frequency is normalized by dividing each frequency value (taken at convenient intervals) by the observed undamped natural frequency, and then comparing this normalized plot with the curves of figure 2, or a similar set of curves. The approximate damping ratio of the rate gyroscope may be determined by taking this to be the damping ratio of the curve with the closest fit to be normalized plot of the recorded data.

4.7.24 Maximum rate sensitivity about spin reference axis test (at zero angular rate about the input reference axis). This test shall measure the pick-off indicated equivalent angular rate about the input reference axis (IRA) due to a level of angular rate applied to the rate gyroscope about the spin reference axis (SRA), while there is no angular rate applied about the input reference axis (IRA). The maximum rate sensitivity about the spin reference axis, under these test conditions, may be determined from this measured value. This test is not intended to precisely measure the actual value of this rate sensitivity; it merely determines that the effective rate sensitivity is less than a certain specified maximum value at a specified level of gyroscope angular rate about the spin reference axis (SRA).

4.7.24.1 Test equipment. A single axis rate table shall be required, which is capable of calibrated angular rotation rates of sufficient magnitude in each direction (CW or CCW) so that the level of angular rotation rate about the spin reference axis (SRA) specified for this test may be attained. A means of accurately determining the table rate (if not an integral part of that equipment) as well as a means of demodulating the pick-off signal (if demodulation is required) shall be provided.

4.7.24.2 Test setup and procedure. Mount the rate gyroscope with the spin reference axis (SRA) parallel to the rate table axis of rotation, which in turn is nominally aligned with the (gravity determined) local vertical. With the rate table stopped (zero angular rate input) operate the gyroscope and pick-off, and record the pick-off (demodulated) signal output. Start the table rotating at a low angular rate in

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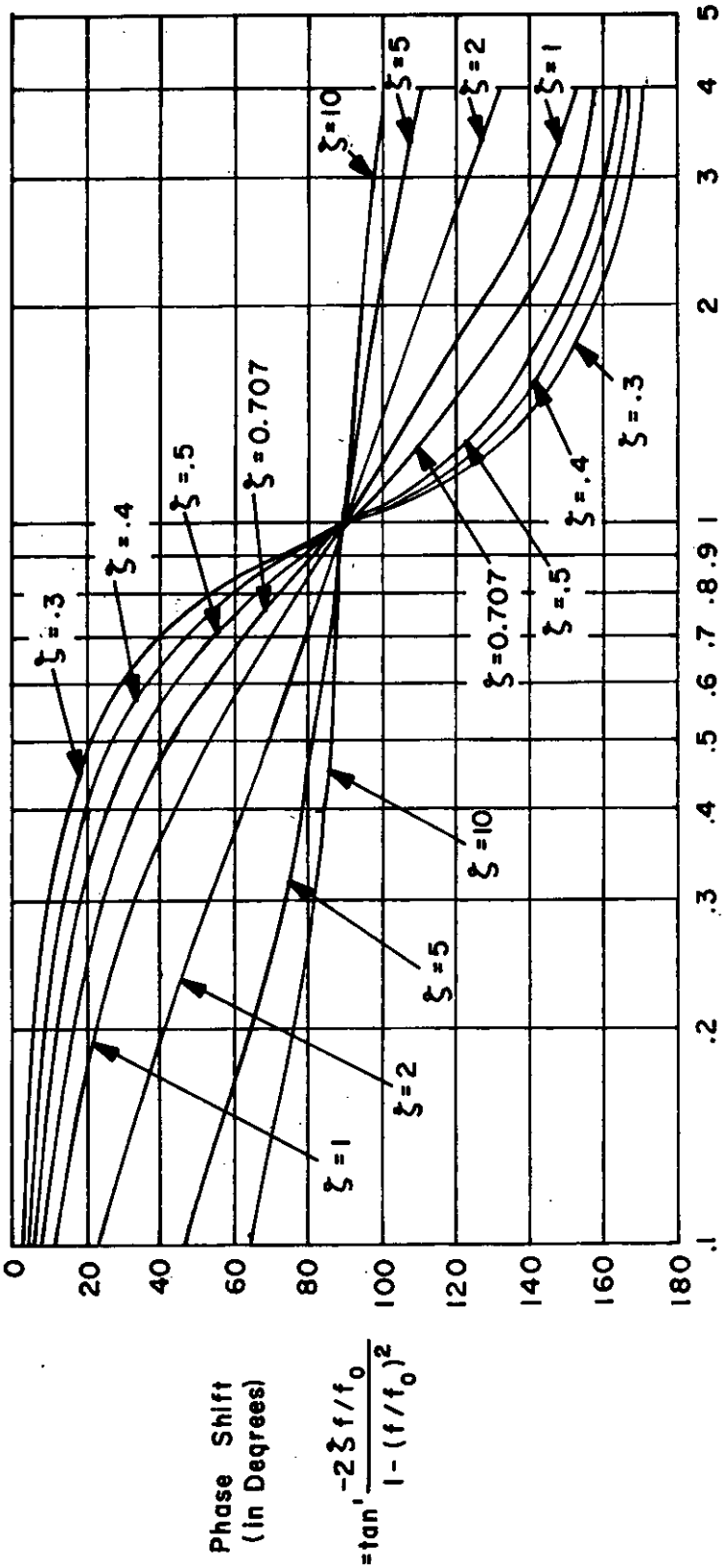


Fig. 2 PHASE RESPONSE OF A SECOND ORDER SYSTEM TO A SINUSOIDAL ACCELERATION (FOR VARIOUS DAMPING RATIOS, ζ)

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one direction, and increase the rotation rate to the required test level of rotation rate about the spin reference axis (SRA) as specified in the detail specification. Record the pick-off signal under this condition. Slow the table to a stop, and then gradually increase the table angular rotation rate in the other direction until again reaching the required level of angular rotation rate about the spin reference axis (SRA). Record the pick-off signal under this condition. Slow the table to a stop, and shut down the rate gyroscope.

* 4.7.24.3 Test results. Take the algebraic sum of the two output signals at the required test level, divide by two, and convert to an equivalent angular rate about the input reference axis using the rate gyro specified scale factor. This maximum observed difference shall be divided by the magnitude of the specified level of angular rotation rate about the spin reference axis (SRA). The result, expressed as a percentage, shall be considered the observed maximum rate sensitivity about the spin reference axis with zero angular rate about the input reference axis.

4.7.25 Maximum rate sensitivity about spin reference axis test (at specified angular rate about the input reference axis). This test shall measure the pick-off indicated equivalent angular rate about the input reference axis (IRA) due to a level of angular rate applied to the rate gyroscope about the spin reference axis (SRA), while there is a specified angular rate applied about the input reference axis (IRA). The observed maximum rate sensitivity about the spin reference axis, under these test conditions, may be determined from this measured value. This test is not intended to precisely measure the actual value of this rate sensitivity; it merely determines that the effective rate sensitivity is less than a certain specified maximum value at a specified level of gyroscope angular rate about the spin reference axis (SRA).

4.7.25.1 Test equipment. A single axis rate table shall be required, which is capable of calibrated angular rotation rates of sufficient magnitude in each direction (CW or CCW) so that the level of total angular rotation rate provides the specified components of angular rate about the spin reference axis (SRA) and the input reference axis (IRA). A means of accurately determining the table rate (if not an integral part of that equipment) shall be provided. A rate gyroscope mounting fixture shall be designed to allow a 90 degree rotation of the rate gyroscope about the principal reference axis which is mutually perpendicular to the input reference axis (IRA)

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and the spin reference axis (SRA). This 90 degree adjustment allows the spin reference axis (SRA) to be positioned at any angle with respect to the table rotation axis; from parallel alignment (with the input reference axis perpendicular to the table axis of rotation) to a position perpendicular to the table axis of rotation (with the input reference axis in parallel alignment with the table axis rotation). This allows the ratio of the angular rate component applied about the spin reference axis (SRA) to the angular rate component applied about the input reference axis (IRA) to be set to any value between zero and an arbitrarily large number. Since the rate table angular rate can also be set, the value of the angular rotation rate about the input reference (IRA) and the value of the angular rotation rate about the spin reference axis (SRA) can be independently determined. A means of demodulating the pick-off signal (if demodulation is required) shall also be provided.

4.7.25.2 Test setup and procedure. Mount the rate gyroscope in the fixture on the rate table so that the gyroscope principal reference axis, mutually perpendicular to the input reference axis (IRA) and the spin reference axis (SRA), is perpendicular to the table axis of rotation. Select both a value for the angle between the spin reference axis (SRA) and the table rotation axis, and an angular rotation rate for the rate table such that the angular rate component about the input reference axis and the angular rate component about the spin reference axis is equal to the values specified in the detail specification. Set the angle to the value selected, and lock the fixture in position. With the rate table stopped, operate the gyroscope and pick-off, and record the pick-off (demodulated) signal. Start the table rotating at a low angular rate in one direction, and increase the rotation rate to a level sufficient to provide the required angular rate components about the input reference axis (IRA) and the spin reference axis (SRA). Record the pick-off signal under this condition. Slow the table to a stop, and then gradually increase the table angular rotation rate in the other direction until reaching the required level of total angular rotation rate. Record the pick-off signal under this condition. Slow the table to a stop, and shut down the rate gyroscope.

4.7.25.3 Test results. For each direction of rotation of the rate table, take the difference between the pick-off signal with the rate table stopped and the pick-off signal with the required conditions of applied angular rotation rate about the spin reference axis (SRA) and the input reference axis (IRA). Take the larger of these differences and convert it to an equivalent angular rotation rate about

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the input reference axis (IRA) using the rate gyroscope specified output function. Subtract from this result the required level of angular rotation rate about the input reference axis (IRA). Divide the remainder by the magnitude of the specified level of angular rotation rate about the spin reference axis (SRA). The final result, expressed as a percentage, shall be considered the observed maximum rate sensitivity about the spin reference axis with a specified angular rate about the input reference axis.

4.7.26 Maximum rate sensitivity about the output axis test. This test shall measure the pick-off indicated equivalent angular rate about the input reference axis (IRA) due to a level of angular rate applied to the rate gyroscope about the output axis (OA), while there is no angular rate applied about either of the other gyroscope principal reference axes (the input reference axis and the spin reference axis). The output axis (OA) is the gyroscope principal reference axis that is mutually perpendicular to the input reference axis (IRA) and the spin reference axis (SRA). The observed maximum rate sensitivity about the output axis (OA), under these test conditions, may be determined from this measured value. This test is not intended to precisely measure the actual value of this rate sensitivity; it merely determines that the effective rate sensitivity is less than a certain specified maximum value at a specified level of gyroscope angular rate about the output axis (OA).

4.7.26.1 Test equipment. A single axis rate table shall be required, which is capable of calibrated angular rotation rates of sufficient magnitude in each direction (CW or CCW) so that the level of angular rotation rate about the output axis (OA) specified for this test may be attained. A means of accurately determining the table rate (if not an integral part of that equipment) as well as a means of demodulating the pick-off signal (if demodulation is required) shall be provided.

4.7.26.2 Test setup and procedure. Mount the rate gyroscope with the output axis (OA) parallel to the rate table axis of rotation, which in turn is nominally aligned with the (gravity determined) local vertical. With the rate table stopped (zero angular rate input) operate the gyroscope and pick-off, and record the pick-off (demodulated) signal. Start the table rotating at a low angular rate in one direction, and increase the rotation rate to the required test level of rotation rate about the output axis (OA), as specified in the detail specification. Record the pick-off signal under this condition. Slow the table to a stop, and then gradually increase the table angular

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rotation rate in the other direction until again reaching the required level of angular rotation rate about the output axis (OA). Record the pick-off signal under this condition. Slow the table to a stop, and shut down the rate gyroscope.

4.7.26.3 Test results. For each of the directions of rotation of the rate table, take the difference between the pick-off signal with the rate table stopped and the pick-off signal with the required magnitude of angular rotation rate about the output axis (OA). Take the larger of these differences and convert it to an equivalent angular rate about the input reference axis (IRA) using the rate gyroscope specified output function. This maximum observed difference shall be divided by the magnitude of the specified level of angular rotation rate about the output axis (OA). The result, expressed as a percentage, shall be considered the observed maximum rate sensitivity about the output axis with zero angular rate about the other gyroscope principal reference axes.

* 4.7.27 Maximum angular acceleration sensitivity about the output axis test. This test shall measure the pick-off indicated equivalent angular rate about the input reference axis (IRA) due to a level of angular acceleration applied to the rate gyroscope about the output axis (OA), while there is no angular rate applied about either of the other gyroscope principal reference axes (input reference axis and the spin reference axis). The output axis (OA) is the gyroscope principal reference axis that is mutually perpendicular to the input reference axis (IRA) and the spin reference axis (SRA). The observed maximum angular acceleration sensitivity about the output axis (OA), under these test conditions, may be determined from this measured value. This test is not intended to precisely measure the actual value of this angular acceleration sensitivity; it merely determines that the effective angular acceleration sensitivity is less than a certain specified level of gyroscope angular acceleration about the output axis (OA).

4.7.27.1 Test equipment. An oscillating rate table with means provided for adjustment of the frequency of the sinusoidal rate over an adequate range, with a sufficient controlled amplitude of sinusoidal angular rate (and consequently a controlled amplitude of angular acceleration), shall be required. The frequency range of the oscillating rate table need extend only to approximately one tenth of the undamped natural frequency of the rate gyroscope under test, but the angular acceleration capability of the oscillating rate table

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for at least one frequency within this frequency range shall be sufficient to provide the peak angular acceleration specified for this test. A means of accurately determining the table angular frequency and any other table parameters necessary to calculate the peak angular acceleration, shall be required. A means of demodulating the pick-off signal (if demodulation is required) shall be provided.

* 4.7.27.2 Test setup and procedure. Mount the rate gyroscope on the test surface of the oscillating rate table with the output axis (OA) parallel to the table axis. With the table stopped, operate the gyroscope and pick-off, and record the pick-off (demodulated) signal. Start the table in oscillation at a chosen frequency less than or equal to one tenth that of the natural frequency of the rate gyroscope under test, and adjust the table dynamic conditions so that the peak angular acceleration of the table is equal to the value specified in the detail specification. Record the sinusoidal waveform of the pick-off signal (this waveform has the same frequency as the table oscillation), taking care to obtain the correct positive and negative peak values. Stop the table oscillation, then shut down the gyroscope.

* 4.7.27.3 Test results. The applied angular acceleration about the output axis (OA) and the pick-off signal waveform, observed under this test condition, shall be considered to be in time phase. Therefore, the observed peak values of pick-off signal can be considered to occur at times of corresponding angular acceleration peaks of the table. From the recorded waveform obtained during table oscillation conditions, select the larger magnitude peak pick-off signal (it may be the positive or the negative peak value), and subtract from it, the pick-off signal with the table stopped. Convert the difference to an equivalent angular rate about the input reference axis (IRA) using the rate gyroscope specified scale factor. Divide this by the magnitude of the peak table angular acceleration (the angular acceleration specified in the detail specification), and the results shall be considered the observed maximum angular acceleration sensitivity about the output axis.

* 4.7.28 Maximum erection error test (vertical gyroscope). This test shall measure the change in the pick-off indicated rotor spin axis position (for each pick-off) with respect to rotation about the output axes, resulting from 180 degree rotation of the vertical gyroscope about the spin reference axis (SRA), when the spin reference axis (SRA) is aligned with the gravity determined local

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vertical. From this data, the observed erection error can be determined in terms of its components for rotation about each of the output axes. This test is not intended to precisely isolate the spin axis erection error by itself, since it does not eliminate the effect of composite error. It merely determines that the effective spin axis erection error is less than a certain specified value.

4.7.28.1 Test equipment. A vertical gyroscope mounting fixture shall be designed to accurately align the input reference axis (IRA) of the vertical gyroscope to the gravity-determined local vertical. The fixture shall also have an axis of rotation accurately aligned with the local vertical. The fixture shall have at least 180 degrees of angular freedom about this axis of rotation. A means of demodulating the pick-off signal for each pick-off (if demodulation is required) shall be provided.

* 4.7.28.2 Test setup and procedure. Mount the vertical gyroscope in the fixture. Operate the vertical gyroscope and its pick-offs, and after allowing a sufficient interval of time for the vertical gyroscope spin axis to erect to its final equilibrium position, record the (demodulated) signal output for each pick-off. Next, in as rapid a sequence as possible, deactivate the spin axis erection system (for both output axes), rotate the vertical gyroscope 180 degrees about the fixture axis of rotation (in either direction sense), and immediately record the new signal output for each pick-off. Shut down the vertical gyroscope.

Rapid performance of this sequence is necessary to minimize the effects of vertical gyroscope drift when the spin axis erection system is deactivated.

* 4.7.28.3 Test results. For each of the output axes (represented by each of the pick-offs), take the difference between the initial pick-off signal and the final pick-off signal obtained immediately after the 180 degree rotation of the vertical gyroscope about the fixture axis of rotation. Divide each of these differences by two, and using the vertical gyroscope specified output function, convert each of these results into an equivalent angular rotation of the rotor spin axis about the gyroscope output axes. The final results, expressed in degrees, are the components of the spin axis erection error for the respective output axes. The magnitude of the observed erection error can be obtained by taking the square root of the sum of the squares of the magnitudes of these components.

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4.7.29 Dynamic drift test (vertical gyroscope). This test shall measure the rotor spin axis free angular drift from the initial erected position (nominally aligned with the spin reference axis and the gravity determined local vertical, or a case determined local vertical), for a gyroscope subjected to the dynamic operating conditions imposed by a two axis sinusoidal motion (Scorsby) table. This shall be measured over an interval throughout which the spin axis erection system is deactivated.

4.7.29.1 Test equipment. A two axis sinusoidal motion (Scorsby) table shall be required. A gyroscope mounting fixture shall be required which positions the spin reference axis (SRA) of the vertical gyroscope in approximate alignment with the gravity determined local vertical when the table is in the reference (leveled) position. A means of demodulating the pick-off signal for each pick-off (if demodulation is required) shall be provided.

* 4.7.29.2 Test setup and procedure. Mount the gyroscope on the table-mounted fixture, and align the spin reference axis (SRA) to an approximate local vertical while the table is stopped in the reference (leveled) position. The table oscillation frequency shall be specified in the detail specification. The amplitude of the table displacement angles shall be such that the rotor spin axis is displaced through at least 50% of the angular freedom ranges. Operate the gyroscope, and after allowing a sufficient interval of time for the vertical gyroscope rotor spin axis to erect to its final equilibrium position, record the (demodulated) signal output for each pick-off. Deactivate the spin axis erection system (for both output axes), then tilt the table so that it is in the correct attitude phase position of the continuous two axis sinusoidal motion cycle. Start the table oscillating and continue for a "reasonable" length of time, such as a typical missile flight time duration. In any case, the time interval shall be the same as that used in the free drift rate test of 4.7.11. At the end of this test interval stop the table motion but allow the gyroscope to continue to run while the table is quickly returned to the reference position, and the signal output of each pick-off is recorded. Shut down the gyroscope in such a way that there is no risk of damage to the gyroscope.

4.7.29.3 Test results. Determine the magnitude deviation of the signal output of each pick-off as the recorded pick-off signal at the end of the test interval minus the pick-off signal recorded at the start of the test interval. When converted to an angle using the

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vertical gyroscope specified output function, the larger of the two values obtained (one value for each pick-off) shall be considered the observed maximum free drift under dynamic conditions. This value shall be within the limit for this parameter as specified without regard to the dynamic oscillation effect.

4.7.30 Torquer test (directional gyroscope). This test shall measure the composite error of the type II (proportional) torquer output torque with respect to the input signal DC voltage, and with nominally zero rotor spin axis displacement from the spin reference axis (SRA). From this data the observed torquer maximum composite error can be determined.

4.7.30.1 Test equipment. A single axis servo-driven turntable shall be required, with a means provided for accurately determining the turntable angular rotation rate about the turntable axis of rotation. A means of demodulating (if demodulation is required) and amplifying the yaw axis pick-off signal to be applied to the turntable servo amplifier input shall be provided. This signal shall be applied in the proper polarity so as to drive the pick-off (demodulated) signal to zero (and maintain the rotor spin axis in nominal alignment with the spin reference axis). Equipment for generating a torquer input signal DC voltage capable of ranging from zero to the maximum input signal DC voltage in either polarity, shall also be required.

* 4.7.30.2 Test setup and procedure. Mount the directional gyroscope on the single axis servo turntable so that the yaw axis is coincident with the servo turntable axis. The servo turntable axis and the coincident output axis of the gyroscope (yaw axis) correspond to the missile yaw axis. Connect the output of the pick-off located on the gyroscope yaw axis to the input of the turntable servo amplifier in the proper polarity to rotate the turntable in the direction which drives this pick-off (demodulated) signal to zero. Connect the torquer input signal source to the input of the torquer. Operate the directional gyroscope, the pick-off, and the associated servo table electronics. After a sufficient interval of operation time for the rotor spin axis to level to some local horizontal equilibrium position, and after the servo table has rotated to a position aligning the rotor spin axis with the spin reference axis, the torquer test cycle may be started. As rapidly as possible (commensurate with the response dynamics of the turntable servo control system, and the necessities of data taking or plotting) increase,

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from initial zero, the torquer input signal DC voltage in the polarity direction causing the turntable to rotate in the positive angular displacement direction of the corresponding missile principal axis (yaw axis). Continue to increase the torquer input signal in this direction until the maximum torquer input signal and the corresponding maximum rate of gyroscope spin axis precession (and identical turntable rotation rate) about the servo turntable axis is reached. Now decrease the torquer input signal through zero (and through the corresponding turntable zero rotation rate) and continue to the opposite voltage polarity until the identical condition is reached in the negative angular displacement direction. Finally, increase the torquer input signal once again until zero torquer input signal and the corresponding zero rotation rate are reached, and stop on that point. Shut down the gyroscope at the end of this test cycle in such a way that there is no risk of damage to the gyroscope.

* 4.7.30.3 Test results. Draw the torquer specified output function line on the data plot. This line is drawn by converting the torquer output torque variable of the torquer specified output function to an equivalent gyroscope angular precession rate using the calculated gyroscope precession scale factor (torque about an output axis to angular rotation rate of the spin axis about the orthogonal output axis). This provides a function of the "ideal" gyroscope angular precession rate vs. the torquer input signal DC voltage. The maximum observed deviation of the precession rate output from this line divided by the sum of the precession rate range (span) expressed as a percentage, shall be considered the observed torquer maximum composite error.

* 4.7.31 Dynamic drift test (directional gyroscope). This test shall measure the rotor spin axis free angular drift about the yaw axis (output axis) from the initial leveled position (nominally aligned with a local horizontal perpendicular to a gravity determined local vertical, or a case determined local horizontal). This shall be measured with the gyroscope subjected to an interval of the dynamic operating conditions imposed by a two axis sinusoidal motion (Scorsby) table, throughout which interval, the spin axis leveling system is deactivated.

4.7.31.1 Test equipment. A two axis sinusoidal motion (Scorsby) table shall be required. A gyroscope mounting fixture shall be required which positions the spin reference axis (SRA) of the directional

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gyroscope in approximate alignment with the local horizontal (a perpendicular to the gravity determined local vertical) when the table is in the reference (leveled) position. A means of demodulating the pick-off signal (if demodulation is required) shall also be provided.

4.7.31.2 Test setup and procedure. Mount the gyroscope on the table-mounted fixture, and align the spin reference axis (SRA) to an approximate local horizontal while the table is stopped in the reference (leveled) position. The table oscillation frequency shall be as specified in the detail specification, and the amplitude of the table displacement angles shall be such that the rotor spin axis is displaced through at least 50% of the angular freedom ranges. Operate the gyroscope, and after allowing a sufficient interval of time for the rotor spin axis to level to some local horizontal equilibrium position, record the pick-off (demodulated) signal. Deactivate the spin axis leveling system, then tilt the table so that it is in the correct attitude phase position of the continuous two axis sinusoidal motion cycle. Start the table oscillating and continue for a "reasonable" length of time, such as a typical missile flight time duration. In any case, the time interval shall be the same as that used in the free drift rate test of 4.7.13. At the end of this test interval stop the table motion but allow the gyroscope to continue to run while the table is quickly returned to the reference position. Record the signal output of each pick-off, and shut down the gyroscope in such a way that there is no risk of damage to the gyroscope.

4.7.31.3 Test results. Determine the deviation of the pick-off signal as the recorded pick-off signal at the end of the test interval minus the pick-off signal recorded at the start of the test interval. When converted to an angle using the directional gyroscope specified output function, the value obtained shall be considered the observed maximum free drift under dynamic conditions. This value shall be within the limit for this parameter as specified without regard to the dynamic oscillation effect.

* 4.7.32 Mechanical overload test (rate integrating gyroscope). The sample rate integrating gyroscope shall be operated while an overload input rotation rate is applied about the input reference axis. The test equipment required and the setup required shall be similar to that of the rate transfer characteristic test for the rate gyroscope (see 4.7.7). No monitoring of the rate integrating gyroscope performance is required during the application of the input overload rates.

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Apply the specified input overload rate first in one direction sense for the specified time interval (see 3.6.5.3), and then likewise in the other direction sense. Finally, subject this sample rate integrating gyroscope to the angular transfer characteristic test of 4.7.14. If the gyroscope then provides satisfactory performance under the conditions of this latter test, it shall be considered to have fulfilled all performance requirements associated with the input overload test.

4.7.33 Dynamic response test (rate integrating gyroscope). This test shall measure the dynamic response of the rate integrating gyroscope to a variable frequency sinusoidal input rate about the input reference axis, while the rate integrating gyroscope is operated as a rate gyroscope. This is done by using an external electronic servo loop incorporating the rate integrating gyroscope torquer, pick-off, and an external servo amplifier, to provide the necessary elastic restraint of the rotor spin axis. From this data, the approximate undamped natural frequency and the approximate damping ratio may be determined; and from these parameters, in turn, the rate integrating gyroscope characteristic time may be calculated. The validity and utility of these parameters are limited in their application to the actual function of this rate integrating gyroscope as a rate gyroscope by the closeness of the approximation of its dynamic response to the well known and tabulated dynamic response of the ideal "second order system" (see 6.3.12).

4.7.33.1 Test equipment. An oscillating rate table shall be required, with means provided for adjustment of the frequency and amplitude of the sinusoidal rate over an adequate range. The frequency range of the oscillating rate table shall extend to a limit well above the expected nominal undamped natural frequency of the rate integrating gyroscope operated as a rate gyroscope. It is possible to adjust the natural frequency of the system (the rate integrating gyroscope operated as a rate gyroscope) by adjusting the gain in the servo amplifier. Thus, the natural frequency may be set to accommodate frequency limitations of the oscillating rate table. The angular rate amplitude should be well up in the input range of the gyroscope, but not so high as to cause overload to occur.

Also required, are a means of accurately determining the frequency and phase of the table input angular rate (if not an integral part of that equipment), a means of demodulating the pick-off signal (if demodulation is required), of amplifying the pick-off (demodulated)

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signal output, and a means of determining the phase difference of the pick-off sinusoidal output signal (or the fundamental component of the demodulated output signal waveform) and the table sinusoidal input rate.

4.7.33.2 Test setup and procedure. Mount the rate integrating gyroscope on the test surface of the oscillating rate table with the input reference axis (IRA) parallel to the table axis. Plot the phase difference of the pick-off (demodulated) signal output and the table input rate as a function of table input frequency. With the table stopped, operate the gyroscope, torquer, pick-off, and servo electronics. Start the table in oscillation at a low frequency and slowly increase the frequency of the table rate input while noting the phase difference which is being recorded. Continue to increase the table input frequency until the phase of the gyroscope pick-off signal lags the phase of the table input rate by 90 degrees. Record the frequency at which this occurs, and continue to increase the frequency above this value until either the limit of the frequency range of the oscillating rate table is reached or the amplitude of the oscillation has decreased to a level where reliable phase readings of the pick-off signal and the table input can no longer be obtained. Stop the oscillating rate table and shut down the system. The slow rate of increase in the frequency of oscillation is to avoid any transient dynamic effects of the rate table or the plotting apparatus, so that the true steady state dynamic response is obtained.

4.7.33.3 Test results. The frequency at which the phase of the pick-off (demodulated) signal output lags the phase of the table input rate by 90 degrees shall be considered as the observed undamped natural frequency. The plot of the phase difference as a function of table rate input frequency is normalized by dividing each frequency value (taken at convenient intervals) by the observed undamped natural frequency, and then comparing this normalized plot with the curves of figure 2, or a similar set of curves. The approximate damping ratio of the rate integrating gyroscope operated as a rate gyroscope may be determined by taking this to be the damping ratio of the curve with the closest fit to the normalized plot of the recorded data. The reciprocal of two times the product of the damping ratio and the undamped natural frequency shall be considered as the approximate characteristic time of the rate integrating gyroscope.

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4.7.34 Maximum angular sensitivity about spin reference axis test (at zero angular displacement about the input reference axis). This test shall measure the pick-off indicated equivalent angular displacement about the input reference axis (IRA) due to an angular displacement applied to the rate integrating gyroscope about the spin reference axis (SRA), while there is no angular displacement applied about the input reference axis (IRA). The observed maximum angular sensitivity about the spin reference axis, under these test conditions, may be determined from this measured value. This test is not intended to precisely measure the actual value of this angular sensitivity; it merely determines that the effective angular sensitivity is less than a certain specified maximum value at a specified level of gyroscope angular displacement about the spin reference axis (SRA).

4.7.34.1 Test equipment. A single axis displacement table shall be required, which is capable of calibrated angular rotation of sufficient magnitude so that the level of angular rotation about the spin reference axis (SRA) specified for this test may be attained. The table should preferably be capable of continuous rotation about its axis or rotation. A means of accurately determining the table angular displacement (if not an integral part of that equipment) as well as a means of demodulating the pick-off signal (if demodulation is required) shall be provided. A means of amplifying the pick-off (demodulated) signal output to be applied to the torquer input so that the rotor spin axis may be driven to the spin reference axis positions shall also be required.

4.7.34.2 Test setup and procedure. Mount the rate integrating gyroscope with the spin reference axis (SRA) parallel to the displacement table axis of rotation. With the displacement table at some convenient reference position, and with arrangements made to amplify the pick-off (demodulated) signal so that it may be connected to the torquer input in the proper polarity to drive the pick-off signal to near zero, operate the gyroscope. Allow a sufficient interval of operation so that dynamic equilibrium is reached with the pick-off output signal at near zero and the gyroscope rotor spin axis nominally aligned with the spin reference axis (SRA). Record the pick-off signal, and then in as rapid a sequence as possible, disconnect the torquer input signal, displace the table (about the table axis of rotation) through an angle specified for this test and record the new pick-off signal. The above angle shall be specified in the detail specification and may be larger than 360 degrees, depending

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on the specified output function. The rapid performance of this sequence is important to reduce the effects of gyroscope drift.

4.7.34.3 Test results. Take the difference between the pick-off signal obtained initially with the torquer input signal applied, and the pick-off signal obtained after the gyroscope was rotated through the specified angular displacement. Convert this difference to an equivalent angular displacement about the input reference axis using the rate integrating gyroscope specified output function. The magnitude of this converted difference shall be divided by the magnitude of the specified angular rotation of the gyroscope about the spin reference axis (SRA). The results, expressed as a percentage, shall be considered the observed maximum angular sensitivity about the spin reference axis with zero angular displacement about the input reference axis.

4.7.35 Maximum angular sensitivity about spin reference axis test (at specified angular displacement about the input reference axis). This test shall measure the pick-off indicated equivalent displacement about the input reference axis (IRA) due to an angular displacement applied to the rate integrating gyroscope about the spin reference axis (SRA), while there is a specified angular displacement applied about the input reference axis (IRA). The observed maximum angular sensitivity about the spin reference axis, under these test conditions, may be determined from this measured value. This test is not intended to precisely measure the actual value of this angular sensitivity; it merely determines that the effective angular sensitivity is less than a certain specified maximum value at a specified level of gyroscope angular displacement about the spin reference axis (SRA).

4.7.35.1 Test equipment. A single axis displacement table shall be required, which is capable of calibrated angular rotation of sufficient magnitude so that the amount of total angular displacement of the test table about its axis of rotation provides the specified components of angular displacement about the spin reference axis (SRA) and the input reference axis (IRA). The table should preferably be capable of continuous rotation about its axis of rotation. A means of accurately determining the table angular displacement (if not an integral part of that equipment) shall be provided. A rate integrating gyroscope mounting fixture shall be designed to allow 90 degree rotation of rate integrating gyroscope about the principal reference axis which is mutually perpendicular to the input reference

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axis (IRA) and the spin reference axis (SRA). This 90 degree adjustment allows the spin reference axis (SRA) to be positioned at any angle with respect to the table rotation axis; from parallel alignment (with the input reference axis perpendicular to the table axis of rotation) to a position perpendicular to the table axis of rotation (with the input reference axis of parallel alignment with the table axis of rotation). This allows the ratio of the angular displacement component applied about the spin reference axis (SRA) to angular displacement component applied about the input reference axis (IRA) to be set to any value between zero and an arbitrary large number. Since the table total angular displacement can also be set, the value of the angular displacement about the input reference axis (IRA) and the value of the angular displacement about the spin reference axis (SRA) can be independently determined. A means of demodulating (if demodulation is required) and amplifying the pick-off signal shall also be provided.

4.7.35.2 Test setup and procedure. Mount the rate integrating gyroscope in the fixture on the displacement table so that the gyroscope principal reference axis, mutually perpendular to the input reference axis (IRA) and the spin reference axis (SRA), is perpendicular to the table axis of rotation. Select both a value for the angle between the spin reference axis (SRA) and the table rotation axis, and a total angular displacement for the table such that the angular displacement component about the input reference axis, and the angular displacement component about the spin reference axis is equal to the values specified in the detail specification. Set the angle to the value selected, and lock the fixture in position. The value required for the total displacement angle may be larger than 360 degrees depending on the rate integrating gyroscope specified output function. With the displacement table at some convenient reference position, and with arrangements made to amplify the pick-off signal so that it may be connected to the torquer input in the proper polarity to drive the pick-off signal to near zero, operate the gyroscope. Allow a sufficient interval of operation so that dynamic equilibrium is reached with the pick-off signal at near zero and the gyroscope rotor spin axis nominally aligned with the spin reference axis (SRA). Record the pick-off signal, and then, in as rapid a sequence as possible, disconnect the torquer input signal, displace the table (about the table axis of rotation) through the total angle required in this test, and record the new pick-off signal. The rapid performance of this sequence is important to reduce the effects of gyroscope drift.

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4.7.35.3 Test results. Take the difference between the pick-off signal obtained initially with the torquer input signal applied, and the pick-off signal obtained after the gyroscope was rotated through the required total angular displacement. Convert this difference to an equivalent angular displacement about the input reference axis using the rate integrating gyroscope specified output function. Now subtract from this result the specified angular displacement about the input reference axis (IRA). Divide the remainder by the magnitude of the specified angular displacement about the spin reference axis (SRA). The results, expressed as a percentage shall be considered the observed maximum angular sensitivity about the spin reference axis with specified angular displacement about the input reference axis.

* 4.7.36 Environmental test (all gyroscope types). Unless otherwise specified in the detail specification each preproduction sample (see 3.7 and 6.2) shall be subjected to the following environmental tests. Prior to and upon completion of all these listed environmental tests, or following the completion of those environmental tests which may be specified by the procuring activity, the sample gyroscope shall be subjected to all the acceptance examinations and tests of 4.5 applicable to the gyroscope type.

a. Operational life test
b. Thermal equilibrium test
c. Thermal shock test
d. Low pressure test
e. Temperature-altitude (pressure) cycling test
f. Vibration test
g. Shock test
h. Acceleration test
i. Humidity test
j. Salt-spray test
k. Sand and dust test

4.7.36.1 Operational life test. The gyroscope shall be subjected to an operational life test determined by the type of gyroscope rotor drive. One test method shall apply for the continuous drive type and another for the drive and coast type.

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4.7.36.1.1 Continuous rotor drive type. For a gyroscope with a continuously driven rotor, the operational life test shall consist of operation of the gyroscope (with all pick-offs energized) on a two axis sinusoidal motion (Scorsby) table, using the oscillation frequency specified in the detail specification. The amplitude of the displacement angles (or the amplitude of the angular rates, as appropriate to the rate gyroscope) shall be such that the rotor spin axis is displaced through at least 50% of the angular freedom ranges (or through at least 50% of the input ranges of the rate or rate integrating gyroscope). Erection and leveling systems of vertical and directional gyroscopes shall be active. The procuring activity may specify (see 6.2) special operational tests for gyroscope auxiliary mechanisms such as caging-uncaging devices, and torquer. For the continuously driven rotor gyroscope, all operation time accumulated by the gyroscope during the acceptance tests applicable to this gyroscope type (see 4.5) may be considered as an initial part of the operational life test time of that gyroscope sample unit. After the total number of operating hours required has been accumulated, the gyroscope shall be subjected to all the acceptance examinations and tests applicable to this gyroscope type (see 4.5). Failure of the gyroscope to meet any of the requirements associated with these tests, or any failure observed during the duration of the life test procedure shall constitute failure of the operational life test.

4.7.36.1.2 Drive and coast rotor drive type. For the gyroscope with a drive and coast rotor, the operational life test shall consist of operation of the gyroscope (with all pick-offs energized) on a two axis sinusoidal motion (Scorsby) table, using the oscillation frequency specified in the detail specification. The amplitude of the displacement angles shall be such that the rotor spin axis is displaced through at least 50% of the angular freedom ranges. The test procedure shall start with the table initially stopped in the reference (leveled) position (displaced from any instantaneous position in its normal displacement cycle). Spin up the gyroscope rotor and energize the pick-offs. Uncage the rotor spin axis, then tilt the gyroscope and table so that it is in the correct attitude phase position of the continuous two axis sinusoidal motion cycle. Start the table oscillating and continue for the specified operation time (see 3.5.4.4.1) of the gyroscope. Shut down the gyroscope and the table in such a way that there is no risk of damage to the gyroscope. Repeat this test a number of times for the same sample unit, until the gyroscope has undergone a total number of operational cycles equal to the number of operational cycles constituting the specified operation life of the gyroscope. The procuring activity may specify

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(see 6.2) special operation tests for gyroscope auxiliary mechanisms such as caging-uncaging devices, and indicated switches. When the total number of operational cycles required by the specified operational life has been attained by the sample gyroscope, it shall be subjected to all the acceptance examinations and tests applicable to this gyroscope type (see 4.5). Failure of the gyroscope to meet any of the requirements associated with these tests or any failure observed during the time interval of performance of the life test operational cycles, shall constitute failure of the operational life test. The operational cycles required for these latter tests shall not be counted or included as part of the operational life requirements.

4.7.36.2 Thermal equilibrium test. The gyroscope shall be subjected to the test procedures detailed in Method 501.1 and Method 502.1 of MIL-STD-810 with the modifications noted in 3.7.2.1. During the interval of each of these tests when the test item is to be operated, the gyroscope shall be subjected to any one of the performance tests (excluding tests a, b, and c) specified under 4.5 applicable to the gyroscope type.

4.7.36.3 Thermal shock test. The gyroscope shall be subjected to the test procedure detailed in Method 503.1 of MIL-STD-810. Additional thermal cycles may be required if the effects of "creep" or mass shifting are suspected to be manifest in gyroscope performance following prolonged storage periods with periodic temperature cycles.

4.7.36.4 Low pressure test. The gyroscope shall be subjected to the test procedure detailed in Method 500.1 (Procedure II) of MIL-STD-810, with the modifications noted in 3.7.2.3. During the interval of this test when it is to be operated, the gyroscope shall be subjected to any one of the performance tests (excluding tests a, b, and c) specified under 4.5 applicable to the gyroscope type.

4.7.36.5 Temperature-altitude (pressure) cycling test. The gyroscope shall be subjected to the test procedure detailed in Method 504.1 of MIL-STD-810, with the modifications noted in 3.7.2.4. During the interval of this test when it is to be operated, the gyroscope shall be subjected to any one of the performance tests (excluding tests a, b, and c) specified under 4.5 applicable to the gyroscope type.

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4.7.36.6 External magnetic fields test. The gyroscope shall be subjected to operation in the presence of magnetic fields of external origin. The gyroscope mounting fixture (and associated performance test equipment such as a single axis rate table or a sidereal table, when required) and the apparatus for producing the magnetic field (Helmholz coils) shall be arranged so that the maximum level of magnetic flux density specified in 3.7.2.5 may be produced in any of the required directions at the center of the gyroscope position, with the gyroscope in any of the required test positions. The required maximum level of D.C. magnetic flux density shall be measured at a point near the center of the gyroscope mounting location, without the gyroscope, mounting fixture or any associated performance test equipment in the mounting proximity. The level of D.C. magnetic flux measured throughout the volume to be occupied by the gyroscope, (under the same conditions of measurement as previous) shall not vary by more than plus or minus 10% from the value specified. The test procedure and performance test requirements shall be determined by the gyroscope type, as specified in the following subparagraphs.

4.7.36.6.1 Type I - free gyroscope. The performance test to be used for the free gyroscope in this test shall be the free drift test of 4.7.5. A modification to the test shall be that the gyroscope shall be mounted in only one position, duplicating the attitude of the gyroscope with respect to the Earth-fixed coordinates used in either one of the two test runs of 4.7.5. With the gyroscope in this one attitude position, the magnetic field flux vector shall be successively aligned with the three gyroscope principal axes for a total of three test runs. During each of the test runs the externally produced magnetic field shall be present during the entire data recording interval. The data shall be analyzed as specified in 4.7.5. The largest observed free drift and the largest observed free drift rate (of the three runs) shall be within the limits specified for these parameters without regard to the magnetic field effects.

4.7.36.6.2 Type II - rate gyroscope. The performance test to be used for the rate gyroscope in this test shall be the tumble test of 4.7.8. The magnetic field flux vector shall be successively aligned with the local horizontal perpendicular to the table rotation axis, and the gravity-determined local vertical for a total of two test runs. During each of the test runs the externally produced magnetic field shall be present during the entire data recording interval. The data shall be analyzed as specified in 4.7.8. The larger observed

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"acceleration-sensitive rate drift" (now actually due to the combined effects of gravity and the magnetic field) of the two runs shall be within the limits specified for this parameter without regard to the magnetic field effect.

4.7.36.6.3 Type III - vertical gyroscope. The performance test to be used for the vertical gyroscope in this test shall be the free drift rate test of 4.7.11. The magnetic field flux vector shall be successively aligned with the three gyroscope principal axes for a total of three test runs. During each of the test runs the externally produced magnetic field shall be present during the entire data recording interval. The data shall be analyzed as specified in 4.6.11. The largest observed free drift and the largest observed free drift rate (of the three runs) shall be within the limits specified for these parameters without regard to the magnetic field effects.

4.7.36.6.4 Type IV - directional gyroscope. The performance to be used for the directional gyroscope in this test shall be the free drift rate test of 4.7.13. The magnetic field flux vector shall be successively aligned with the three gyroscope principal axes for a total of three test runs. During each of the test runs the externally produced magnetic field shall be present during the entire data recording interval. The data shall be analyzed as specified in 4.7.13. The largest observed free drift and the largest observed free drift rate (of the three runs) shall be within the limits specified for these parameters without regard to the magnetic field effects.

* 4.7.36.6.5 Type V - rate integrating gyroscope. The performance test to be used for the rate integrating gyroscope in this test shall be the tumble test of 4.7.16. The magnetic field flux vector shall be successively aligned with the local horizontal perpendicular to the table rotation axis, and the gravity-determined local vertical for a total of two test runs. During each of the test runs the externally produced magnetic field shall be present during the entire data recording interval. The data shall be analyzed as specified in 4.7.16, with the modification that only the fundamental component proportional to systematic "acceleration-sensitive drift rate" (now actually due to the combined effects of gravity and the magnetic field) is calculated. The final result of the observed "maximum acceleration-sensitive drift rate" shall be within the limits specified for this parameter without regard to the magnetic field effect.

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4.7.36.7 Radiation test. The test method to be used for the radiation exposure of the sample gyroscope shall be as specified in the detail specification. Factors to be considered shall include the availability and types of radiation sources applicable for gyroscope test purposes, and the feasibility of gyroscope operational tests during the radiation exposure. Following the specified radiation exposure the gyroscope shall be subjected to the insulation resistance test of 4.7.2, and the measured values shall be within the limits specified for this parameter without regard to the radiation effects. For the Type III and Type IV (vertical and directional) gyroscopes, an operation test with the erection and leveling systems active, as in the angular transfer characteristic tests of 4.7.10 and 4.7.12 may be required to check for radiation damage to electronic active and passive devices which may be included in the gyroscope as part of the erection and leveling servo loops.

4.7.36.8 Vibration test. The gyroscope shall be subjected to the test procedure detailed in Method 514.1 of MIL-STD-810. During the intervals of this test when it is to be operated, the gyroscope shall be subjected to a performance test as determined by the gyroscope type, and as specified in the following subparagraphs.

4.7.36.8.1 Type I - free gyroscope. The performance test to be used for the free gyroscope in this test shall be the free drift test of 4.7.5.

4.7.36.8.2 Type II - rate gyroscope. The performance test to be used for the rate gyroscope in this test shall be the vibration test of 4.7.9. A modification to the test shall be that the environmental amplitude and frequency of vibration as required by Method 514.1 of MIL-STD-810 shall be used in the procedure of this test instead of the values specified in the vibration test of 4.7.9 for the purpose of measuring the acceleration-squared sensitive rate drift.

4.7.36.8.3 Type III - vertical gyroscope. The performance test to be used for the vertical gyroscope in this test shall be the free drift rate test of 4.7.11.

4.7.36.8.4 Type IV - directional gyroscope. The performance test to be used for the directional gyroscope in this test shall be the free drift rate test of 4.7.13.

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* 4.7.36.8.5 Type V - rate integrating gyroscope. The performance test to be used for the rate integrating gyroscope in this test shall be the free drift test of 4.7.15. Modifications to the test shall be that during the calculation of the test results, the average value of the torquer signal shall be the only data for consideration. When this is converted to an equivalent drift rate using the torquer design scale factor (signal to torque) and the calculated gyroscope precession scale factor (torque about the output axis or output axis to angular rotation rate of the spin axis about the input reference axis), the result shall be within the limits specified for this test in the detail specification. The random drift rate data shall not be considered.

4.7.36.9 Shock test. The gyroscope shall be subjected to the shock test detailed in Method 516.1 of MIL-STD-810. Either Procedure I, Procedure IV, or Procedure V, as specified in the detail specification, shall be used. The detail specification shall also specify if the gyroscope is to be operating or not operating during the shock test, and if the gyroscope is to be caged or uncaged during either of the prior conditions (if applicable to the gyroscope type under consideration). If operation is required during the shock test, no performance monitoring is necessary. Consideration shall be given to the possibility of requiring distinctly different shock tests for each of the gyroscope (operating and not operating, caged and uncaged) conditions, in order to more accurately simulate actual shock environments for drive and coast rotor drive gyroscopes.

* 4.7.36.10 Acceleration test. When required by the procuring activity (see 6.2), the gyroscope shall be subjected to the acceleration test detailed in Method 513.1 of MIL-STD-810. Procedure II shall be used, and the test method shall be by rectilinear acceleration as in a rocket sled test, rather than by centripetal acceleration as in a centrifuge test. The gyroscope shall be subjected to a performance test as determined by the gyroscope type, and as specified in the following subparagraphs.

4.7.36.10.1 Type I - free gyroscope. The performance test to be used for the free gyroscope in this test shall be the free drift test of 4.7.5.

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4.7.36.10.2 Type II - rate gyroscope. The performance test to be used for the rate gyroscope shall consist of operation of the gyroscope with the unit rigidly mounted in the test vehicle in the five successive orientations required by the Method 513.1 of MIL-STD-810, with monitoring and recording of the pick-off (demodulated) signal output for each of the data runs. The five successive mounting orientations are required to align the three gyroscope principal axes with the axis of test vehicle rectilinear acceleration (in the required direction), such that the specified levels of acceleration along each of the corresponding missile principal axes (and in the specified direction of the axes) are applied to the gyroscope. Select the largest recorded value of the pick-off output from among the test data of all of the runs. Convert it to the equivalent angular input rate using the rate gyroscope specified output function. Then divide it by the acceleration in g's of the test run from which it was obtained. The result shall be within the limit specified for the rate gyroscope acceleration-sensitive rate drift.

4.7.36.10.3 Type III - vertical gyroscope. The performance test to be used for the vertical gyroscope in this test shall be the free drift rate test of 4.7.11. In the Method 513.1 of MIL-STD-810, the directions of applied acceleration in this test series shall be "fore", "aft", and "lateral" only, with "up" and "down" excluded.

4.7.36.10.4 Type IV - directional gyroscope. The performance test to be used for the directional gyroscope in this test shall be the free drift rate test of 4.7.13. In the Method 513.1 of MIL-STD-810, the directions of applied acceleration in this test series shall be "fore", "aft", and "lateral" only, with "up" and "down" excluded.

4.7.36.10.5 Type V - rate integrating gyroscope. The performance test to be used for the rate integrating gyroscope shall consist of operation of the gyroscope with the unit rigidly mounted in the test vehicle in the five successive orientations required by the Method 513.1 of MIL-STD-810, with monitoring and recording of the pick-off (demodulated) signal output for each of the data runs. The five successive mounting orientations are required to align the three gyroscope principal axes with the axis of test vehicle rectilinear acceleration (in the required direction), such that the specified levels of acceleration along each of the corresponding missile principal axes (and in the specified direction of these

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axes) are applied to the gyroscope. The pick-off (demodulated) signal output is amplified and connected to the torquer input in the proper polarity to drive the pick-off output signal to zero. This shall be done prior to the application of any rectilinear acceleration or the recording of the pick-off signal output data. After a sufficient time interval of operation has elapsed so that the pick-off signal is observed to be driven to nominal zero, deactivate the torquer, begin the pick-off signal data recording, and apply the specified rectilinear acceleration. Select the largest recorded value of the pick-off output from the test data of all of the runs. Convert it to the equivalent input drift angle using the gyroscope specified output function and the pick-off design scale factor. When the result is divided by the acceleration in g's of the test run from which it was obtained, and also divided by the time elapsed from the start of the application of the acceleration to the time at which this largest value was recorded, the final result shall be within the limit specified for the rate integrating gyroscope acceleration-sensitive drift rate.

4.7.36.11 Humidity test. The gyroscope shall be subjected to the test procedure detailed in Method 507.1 of MIL-STD-810. Immediately after the exposure of the test item to the humidity cycles, and at the conclusion of the test when operation of the test item is required, the gyroscope shall be subjected instead to the insulation resistance test of 4.7.2.

4.7.36.12 Fungus test. The gyroscope or its constituent materials shall be subjected to the test procedure detailed in Method 508.1 of MIL-STD-810.

4.7.36.13 Salt-spray test. The gyroscope shall be subjected to the test procedure detailed in Method 509.1 of MIL-STD-810. At the conclusion of the test when operation of the test item is required, the gyroscope shall be subjected instead to the insulation resistance test of 4.7.2.

4.7.36.14 Sand and dust test. The gyroscope shall be subjected to the test procedure detailed in Method 510.1 of MIL-STD-810. At the conclusion of the test when operation of the test item is required, the gyroscope shall be subjected to any one of the performance tests (excluding tests a, b, and c) specified under 4.5 applicable to the gyroscope type.

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5. PREPARATION FOR DELIVERY

* 5.1 Preservation and packaging. Preservation and packaging of the gyroscope shall be as specified in the detail specification.

5.2 Marking. Marking of the packaged gyroscope shall be in accordance with MIL-STD-129 as applicable.

6. NOTES

6.1 Intended use. Gyroscopes of the types specified herein are intended for use as components of the guidance and control systems of command guided missiles.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this publication

b. When preproduction tests are not required (see 3.1 and 4.4)

c. Number of preproduction samples required (see 4.4.1)

d. Electrical connector(s) to be used (see 3.4.4)

e. Special surface treatments for metal parts (see 3.4.5)

f. Applicability of corrosive fumes requirement (see 3.4.7)

g. Applicability of explosive atmosphere requirement (see 3.4.8)

* h. Limitations on use of auxiliary units for external gain modification (see 3.5.8)

i. Applicability of maximum threshold, and maximum resolution requirements (see 3.6.2.6.3 and 3.6.5.7.2)

j. Deviations from, or exceptions to, requirements for rate sensitivity about the spin reference axis (see 3.6.2.6.8)

k. Deviations from, or exceptions to, requirements for angular sensitivity about the spin reference axis (see 3.6.5.7.8)

l. When caging is required for a vertical gyroscope (see 3.6.3.5)

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m. When a pitch axis torquer is required for a directional gyroscope (see 3.6.4.3)

n. Pick-off type, if type II pick-off is not desired for rate integrating gyroscope (see 3.6.5.4)

o. Exceptions to specified environmental conditions and tests (see 3.7 and 4.7.36)

p. When gyroscope is to be shipped as part of a larger assembly (see 3.7.1)

q. Applicability of external magnetic fields requirement (see 3.7.2.5)

r. Alternate requirements for shock environment (see 3.7.2.8)

s. When hermetic sealing is required (see 3.7.3)

t. Applicability of electromagnetic interference requirement (see 3.11)

u. Deviations from the specified test conditions (see 4.3)

v. Exceptions to specified acceptance examinations and test (see 4.5)

w. Exceptions to specified periodic production tests (see 4.6)

* x. Special operational tests for gyroscopes' auxiliary mechanisms

y. Applicability of acceleration requirement and tests (see 3.7.2.9 and 4.7.36.10)

z. Alternate radiation exposure and test procedures if required (see 3.7.2.6)

* 6.3 Definitions. All definitions applicable to gyroscopes, including those specified herein, shall be in accordance with Aerospace Industries Association EETC Report No. 5 (Revision A)(see 2.2).

6.3.1 Missile principal reference axes. These are the three mutually perpendicular missile body-fixed axes of rotation of the missile used to describe missile attitude and changes in attitude. They are depicted in Figure 1. Missile changes in attitude are described as angular displacements of the missile body about these axes, and the polarities of these changes in missile attitude for each of the three missile principal reference axes are also shown in Figure 1.

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* 6.3.2 Gyroscope principal reference axes. These are the three mutually perpendicular axes of the gyroscope structure used to describe gyroscope rotor attitude and changes in attitude with respect to the stationary portions of the gyroscope structure.

* 6.3.2.1 Spin reference axis (SRA). The SRA is an axis normal to the input reference axis and is nominally parallel to the spin axis when the gyro outputs have specified values, usually null.

* 6.3.2.2 Output axis. An output axis is an axis-of-freedom provided with a pick-off which generates an output signal as a function of the output angle.

* 6.3.3 Gimbal freedom. Gimbal freedom is the maximum angular-displacement of a gimbal about its axis and is measured in degrees.

6.3.4 Angular freedom span. This is the sum of the angular freedom ranges taken without regard to the polarities (the sum of the range magnitudes or absolute values). It is measured in degrees.

* 6.3.5 Input range. Input range is the absolute value of the difference between zero input and an input limit.

* 6.3.6 Zero offset. Zero offset is the gyro output when the input rate is zero, generally expressed as an equivalent input rate.

* 6.3.7 Pick-off nominal zero displacement angle. For a gyroscope rotor spin axis rotation about a given perpendicular output axis, this is the angular position defined by the spin axis at the occurrence of the minimum signal output of the pick-off, or "cross-over" of the pick-off output signal polarity.

* 6.3.8 Gimbal error. Gimbal error is the error resulting from angular displacements of gimbals from their reference positions such that gimbal pick-offs do not measure the true angular motion of the case about the input reference axis.

* 6.3.9 Input reference axis (IRA). An input reference axis is the direction of an axis as defined by the case mounting surfaces and/or external case markings. It is nominally parallel to an input axis and is normal to the spin reference axis.

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* 6.3.10 Gyro scale factor. Scale factor is the ratio of a change in output to a change in the input intended to be measured. Scale factor is generally evaluated as the slope of the straight line that can be fitted by the method of least squares to input-output data obtained by varying the input cyclically over the input span.

* 6.3.11 Torquer scale factor. This is the desired "ideal" linear input-output relationship of the nominally linear transducer of input electrical signal to output torque, as in the proportional Type II torquer of this specification.

6.3.12 Undamped natural frequency. The "undamped natural frequency" or "natural frequency" of a rate gyroscope is defined as being that frequency at which the gyroscope output lags the gyroscope input by 90 degrees in time phase. A unique undamped natural frequency exists only for the ideal linear second order system, and therefore the validity of any measured "undamped natural frequency" of a rate gyroscope (the only type which has an approximate second order system response) is strictly dependent on the closeness of this second order approximation.

6.3.13 Damping ratio. The "damping ratio" of a rate gyroscope is defined as the ratio of actual damping to critical damping for the ideal linear second order system. The "damping ratio" of a rate gyroscope is found by comparing the normalized measured dynamic response characteristic curve with the set of dynamic characteristic curves of an ideal linear second order system for corresponding excitation conditions, where each characteristic curve of the set corresponds to a different value of the damping ratio. The damping ratio of the matching dynamic response characteristic curve is called the rate gyroscope "damping ratio". The validity of any measured "damping ratio" of a rate gyroscope, is strictly dependent on its approximation to a linear second order system.

* 6.3.14 Drift rate. Drift rate is the time rate of change of the angular deviation of the spin axis from the desired orientation in space. It consists of random and systematic drift rate components.

6.3.15 Definition of gyroscope types. The nature of each of the types of gyroscopes listed 1, 2 is defined in the following respective paragraphs.

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* 6.3.15.1 Type I - free gyroscope. A free gyro is a two-degree-of-freedom gyro in which the spin axis may be oriented in any specified attitude. In this gyro, output signals are produced by an angular displacement of the case about an axis other than the spin axis.

* 6.3.15.2 Type II - rate gyroscope. A rate gyro is generally a single-degree-of-freedom gyro having primarily elastic restraint of the spin axis about the output axis. In this gyro, an output signal is produced by precession of the gimbal, which is proportional to the angular rate of the case about the input axis.

* 6.3.15.3 Type III - vertical gyroscope. A vertical gyro is a two-degree-of-freedom gyro with provision for maintaining the spin axis vertical. In this gyro, output signals are produced by gimbal angular displacements which correspond to angular displacements of the case about two nominally orthogonal, horizontal axes.

* 6.3.15.4 Type IV - directional gyroscope. A directional gyro is a two-degree-of-freedom gyro with provision for maintaining the spin axis approximately horizontal. In this gyro an output signal is produced by gimbal angular displacement which corresponds to the angular displacement of the case about an axis which is nominally vertical.

* 6.3.15.5 Type V - rate integrating gyroscope. A rate-integrating gyroscope is a single-degree-of-freedom gyro having primarily viscous restraint of the spin axis about the output axis. In this gyro an output signal is produced by precession of the gimbal, which is proportional to the integral of the angular rate of the case about the input axis.

* 6.3.16 Composite error. Composite error is the maximum deviation of the output data from the specified output function.

* 6.3.17 Nutation. Nutation is the oscillation of the spin axis of a two-degree-of-freedom gyro about two orthogonal axes normal to the mean position of the spin axis.

* 6.3.18 Systematic drift rate. Systematic drift rate is that component of drift rate that is correlated with specific operating conditions. It is composed of acceleration-sensitive drift rate and acceleration-insensitive drift rate. It is expressed as angular displacement per unit time.

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* 6.3.19 Erection. Erection is the process of aligning, by precession, a reference axis with respect to the vertical.

* 6.3.20 Caging. Caging is the process of orienting and mechanically locking one or more gyro axes or gimbal axes to a reference position.

* 6.3.21 Changes to previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MI
Navy - AS
Air Force - 19

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Air Force - 19

User Activities:
Navy - MC
Air Force - 11, 15, 44, 70, 71

Preparing Activity:
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SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO (If "yes", in what way?)

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity - Optional) | DATE

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